Safe and Agile Transportation of Cable-Suspended Payload via Multiple Aerial Robots

Yongchao Wang, Junjie Wang, Xiaobin Zhou, *Member, IEEE*, Tiankai Yang, Xin Zhou, Chao Xu, *Senior Member, IEEE*, and Fei Gao, *Member, IEEE*

***Abstract*—Transporting a heavy payload using multiple aerial robots (MARs) is an efficient manner to extend the load capacity of a single aerial robot. However, existing planning schemes for the MAR transportation system (MARTS) still lack the capability to generate a collision-free and dynamically feasible trajectory in real time. Therefore, they are limited to low-agility transportation in simple environments. To bridge the gap, we propose a complete planning scheme for the MARTS, achieving safe and agile aerial transportation (SAAT) of a cable-suspended payload in complex environments. Flatness map for the motor's revolutions per minute of the aerial robot, considering the complete kinematic constraint and the dynamical coupling between each aerial robot and payload, is derived. To improve the responsiveness for the generation of the safe, dynamically feasible, and agile trajectory in complex environments, a real-time spatiotemporal trajectory planning scheme is proposed for the MARTS. Besides, we break away from the reliance on the state measurement for both the payload and cable, as well as the closed-loop control for the payload, and integrate a fully distributed control scheme to track the agile trajectory that is robust against imprecise payload mass, nonpoint mass payload, wind disturbances, and communication delays. The proposed schemes are extensively validated through benchmark comparisons, ablation studies, and simulations. Finally, extensive real-world experiments are conducted on practical MARTSs containing different numbers of aerial robots with onboard computers and sensors. The result validates the efficiency and robustness of our proposed schemes for the SAAT in complex environments.**

***Index Terms*—Aerial transportation, distributed robust control, multiple aerial robots (MARs), trajectory planning.**

Received 18 August 2025; revised 29 January 2026; accepted 18 February 2026. Date of publication 24 March 2026; date of current version 22 April 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62403419, and Grant 62303412 and in part by the National Key R&D Program of China under Grant 2023YFB4706600. This article was recommended for publication by Associate Editor C. Verginis and Editor M. Schwager upon evaluation of the reviewers' comments. *(Yongchao Wang and Junjie Wang contributed equally to this work.) (Corresponding author: Fei Gao.)*

Yongchao Wang is with the School of Aeronautic Science and Engineering, Beihang University, Beijing 100191, China (e-mail: wangyongchao@buaa.edu.cn).

Junjie Wang, Tiankai Yang, and Chao Xu are with the Key Laboratory of Industrial Control Technology, Institute of Cyber-Systems and Control, Zhejiang University, Hangzhou 310027, China (e-mail: wwwangjunjie@zju.edu.cn; tkyang@zju.edu.cn; cxu@zju.edu.cn).

Xiaobin Zhou is with the School of Robotics and Automation, Nanjing University, Suzhou 215163, China (e-mail: xiaobin_nju@nju.edu.cn).

Xin Zhou is with the Differential Robotics Technology Company, Hangzhou 311100, China (e-mail: iszhouxin@zju.edu.cn).

Fei Gao is with the Key Laboratory of Industrial Control Technology, Institute of Cyber-Systems and Control, Zhejiang University, Hangzhou 310027, China, and also with the Differential Robotics Technology Company, Hangzhou 311100, China (e-mail: fgaoaa@zju.edu.cn).

Digital Object Identifier 10.1109/TRO.2026.3677010

Symbol Definition

$\mathcal{I}, \mathcal{B}_n$	World frame, the nth robot's body frame.
$\mathbf{z}_{\mathcal{I}}, \mathbf{z}_{\mathcal{B}}^n \in \mathbb{R}^3$	z-axes of $\mathcal{I}$ and $\mathcal{B}_n$.
$\mathbf{e}_3$	Unit vector $[0, 0, 1]^T$.
$m_L \in \mathbb{R}_{>0}$	Mass of payload.
$m_n \in \mathbb{R}_{>0}$	Mass of the nth aerial robot.
$\mathbf{J}_n \in \mathbb{R}^{3\times 3}$	Rotational inertia of the nth aerial robot.
$\mathbf{p}, \mathbf{v} \in \mathbb{R}^3$	Position and velocity of payload.
$\mathbf{p}_n, \mathbf{v}_n \in \mathbb{R}^3$	Position and velocity of the nth aerial robot.
$\mathbf{p}_n^k \in \mathbb{R}^3$	Position of the kth sampled point on the nth cable.
$\mathbf{R}_n \in \mathrm{SO}(3)$	Rotation matrix of the nth aerial robot.
$\psi_n \in \mathbb{R}$	Yaw angle of the nth aerial robot.
$\mathbf{q}_n$	Quaternion of the nth aerial robot.
$\boldsymbol{\omega}_n \in \mathbb{R}^3$	Body rate w.r.t. $\mathcal{B}_n$ of the nth aerial robot.
$\boldsymbol{\rho}_n \in S^2$	Unit vector of the nth cable w.r.t. $\mathcal{I}$, pointing from payload to the nth aerial robot.
$\mathbf{M} \in \mathbb{R}^{4\times 4}$	Mixing matrix for the aerial robot.
$\boldsymbol{\tau}_n \in \mathbb{R}^3$	Control moment of the nth aerial robot w.r.t. $\mathcal{B}_n$.
$\mathbf{f}_n \in \mathbb{R}^3$	Mass-normalized thrust vector of the nth aerial robot.
$f_n \in \mathbb{R}_{>0}$	Mass-normalized thrust of the nth aerial robot.
$F_n \in \mathbb{R}_{>0}$	Tension of the nth cable.
$\mathbf{F}_n \in \mathbb{R}^3$	Force vector exerted by the nth cable on the payload.
$\boldsymbol{\zeta}_n \in \mathbb{R}^3$	Vector pointing from the center of mass of the nth aerial robot to the tethering point of the nth cable.
$r_{n,i} \in \mathbb{R}_{>0}$	RPM of the nth aerial robot's ith motor.
$\mathbf{r}_n \in \mathbb{R}_{>0}^4$	Vector denoting $[r_{n,1}, r_{n,2}, r_{n,3}, r_{n,4}]^T$.
$\theta_n, \phi_n \in \mathbb{R}$	Pitch angle, azimuth angle of the n^{th} cable w.r.t. $\mathcal{I}$.
$l \in \mathbb{R}_{>0}$	Length of the nth cable.
$g \in \mathbb{R}_{>0}$	Gravitational acceleration.

I. Introduction

Aerial transportation is more time-efficient compared to ground transportation, as it is unaffected by terrain and traffic congestion. In areas such as logistics, medical rescue, and dangerous fire and disaster scenes, various time-sensitive supplies and equipment can be delivered by aerial transportation. Transporting a payload via a single aerial robot has been widely studied [1], [2], [3], [4], [5], [6], [7], [8], [9]. However, the limited load capacity of a single aerial robot greatly restricts its application in different fields.

Using multiple aerial robots (MARs) to transport a payload collaboratively can enhance the load capability of an aerial robot of arbitrary size. The payload can be attached to the MARs via rigid connections such as grippers [10], electromagnets [11], and rigid rods [12], making up a large underactuated system whose acceleration can only be provided by first regulating the attitude of the whole system, resulting in the loss of the system's agility as the payload increases the system's rotational inertia. Therefore, using cables to suspend the payload has attracted extensive interest from the robotics community [13], [14], [15], [16], [17] and is preferred as it not only eliminates the need for additional actuators and reduces structural complexity, thus increasing the load capacity of the system, but also allows agile transportation while minimizing the payload's attitude. In this article, we consider the cable-suspended payload transportation problem via the MAR transportation system (MARTS).

Speedy transportation of a payload in complex environments relies on a planner to generate a safe and agile trajectory. However, this is not an easy task since the dynamical coupling and the kinematic constraint between each aerial robot and payload introduced by the tight cable complicate the problem. Several practical challenges are analyzed as follows.

The first challenge is how to ensure the *dynamical feasibility* of the agile trajectory planned for the MARTS. Due to the limited battery life, increasing the agility of the trajectory is necessary to improve the mission efficiency [18]. However, as the trajectory becomes more agile, the motors' revolutions per minute (RPMs) of the aerial robot may transiently or persistently exceed the upper limit that the voltage of the battery can tolerate, causing large tracking errors and thus resulting in system divergence and even crashes. In particular, the dynamical coupling can exacerbate this trend. To ensure the dynamical feasibility, delicate constraints need to be enforced on the motors' RPMs. However, the kinematic constraint severely complicates the flatness map to calculate the motor's RPM. Therefore, flatness maps fully considering the dynamical coupling and the kinematic constraint existing in the MARTS need to be derived to calculate the motors' RPMs of each aerial robot.

The second challenge is the responsiveness of the planner to generate safe and agile trajectories. The collision avoidance of the MARTS is crucial, as collisions involving the entire system with obstacles or reciprocal collisions between aerial robots can lead to crashes. For the speedy transportation of the MARTS in complex and unknown environments, high-frequency replanning is needed to adapt to rapid changes in the local environment and ensure that collisions can be avoided. However, numbers of both the variables and the constraints used for the trajectory optimization are approximately proportional to the number of aerial robots contained in the MARTS, which increases the solving complexity of the trajectory optimization problem and adversely affects the responsiveness of the planner, especially for a MARTS with limited onboard resources. Thus, the responsiveness of the planner is also important as rapid trajectory generation enhances the mission efficiency and improves the system's ability to react to sudden changes, such as target alterations.

Based on the above analysis, we propose complete planning and control schemes for the MARTS. To the extent of our knowledge, this is the first work that achieves the real-time trajectory generation for the safe and agile aerial transportation (SAAT) of cable-suspended payloads by the MARTS in complex environments and achieves the transportation of a payload approaching the maximum agility of the MARTS by pushing the aerial robot's thrust to its limit. In this work, we indirectly represent the aerial robot's position via the cable's directional vector and the payload's position, such that the kinematic constraint can be eliminated. First, we explicitly derive the flatness maps based on this indirect position representation to obtain the motors' RPMs of the aerial robot, considering the dynamical coupling with the payload. Second, we propose a lightweight centralized trajectory planning scheme to quickly generate a trajectory based on a trajectory class represented by sparse spatiotemporal parameters. This planning scheme constructs not only the obstacle avoidance constraint for each component of MARTS and the reciprocal avoidance constraint for each pair of aerial robots to ensure the safety of the trajectory, but also the delicate dynamical feasibility constraints to ensure that the agile trajectory is dynamically executable. Besides, it also constructs vectorial constraints for each cable's force, which are dexterously eliminated by diffeomorphisms to avoid the undesirable local-optimum trajectory. Third, we integrate a fully distributed control scheme to track the agile trajectory, which gets rid of the reliance on the state measurements for both payload and cable and is robust against the imprecise estimation of payload's mass, nonpoint mass payloads, wind disturbances, and communication delays. Finally, we deploy our planning and control schemes on practical MARTSs containing different numbers of aerial robots and design various experiments to verify our schemes. Partial results are shown in Fig. 1. Contributions of this article are listed as follows.

1) Flatness maps that consider the additional force and torque introduced by the dynamical coupling with the payload are derived based on the indirect position representation for the motors' RPMs of the aerial robot.
2) We present a real-time trajectory planner for the MARTS, which considers the dynamical feasibility of the aerial robot, the collision avoidance of the entire system in complex environments, and the finite vectorial ranges of the cables, to generate safe and agile trajectories.
3) An autonomous MARTS is integrated with a robust distributed control scheme. Moreover, we open-source the codes for both the planner and the controller to facilitate further development of the MARTS by the community.
4) A variety of simulations and real-world experiments that validate the effectiveness of the proposed planning and control schemes on practical MARTSs are presented.

II. Related Works

A. Control for MARTS

The leader–follower control schemes for the transportation of a beam-like payload by two aerial robots are studied in [19], [20], [21]. However, the follower's controller can only react to the

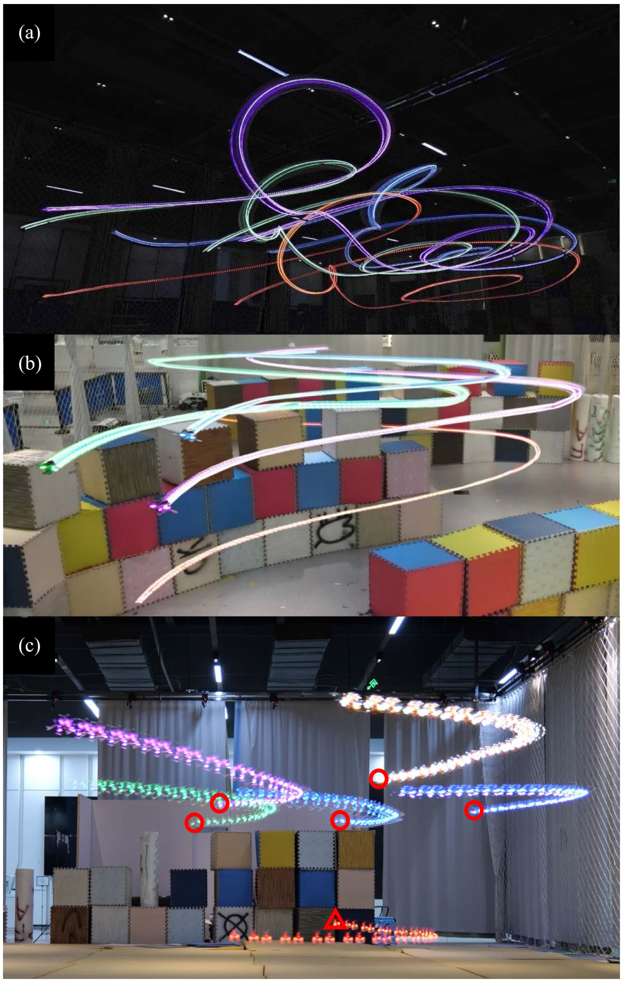


Fig. 1. Simulations and real-world experiments of our MARTS. (a) Agile transportation in free space, approaching the limit of thrust that can be provided by the practical aerial robot in the MARTS. (b) SAAT in a complex environment. (c) Agile transportation by the MARTS containing five aerial robots. Watch our attached videos for more information at https://youtu.be/2Xc8gQEhRxU.

leader's motion passively, which leads to a nonsmooth trajectory. Geometric controllers are studied in [22], [23], and [24] to help the MARTS track a trajectory planned just for the payload. However, they cannot guarantee the avoidance of reciprocal collision between aerial robots since they use a determinate minimum-norm principle to allocate each cable's desired force vector. To ensure reciprocal avoidance, optimization-based [25] and nonlinear model-predictive control (NMPC)-based [26] force allocation methods are proposed via exploiting redundancy in the null space to ensure the minimum safety distance between aerial robots. However, the controllers [22], [23], [24], [25], [26] rely on special sensors or marks mounted on the payload to estimate or measure the payload's state, which is not convenient in practical deployment. These controllers can be classified as centralized, as all force allocations are centralized. The time delay introduced by this centralized computation and data transmission will limit the control performance, especially for agile transportation. Wahba and Hönig [27] propose a distributed force allocation method, in which each aerial robot can update its desired force vector independently via optimizing the same problem with only one global minima, such that consistent results can be ensured. However, this method also does not fundamentally address the dependence on the payload's state. A common denominator of the controllers [22], [23], [24], [25], [26], [27] is that they pursue closed-loop control for the payload. However, the oscillation existing in the payload's tracking error will directly lead to the oscillation of the aerial robot's desired position. Moreover, the control force for the payload can only be provided indirectly once the aerial robot has adjusted its attitude and thrust to regulate the cable's tension vector. Therefore, the response lag and the inevitable control error will fundamentally restrict the performance of these controllers.

To cope with the drawbacks of these controllers mentioned above, several fully distributed control schemes with observers to estimate and compensate for the cable's force are proposed [28], [29] to support closed-loop formation control for each aerial robot. Nevertheless, agile transportation is still rarely seen in existing works. The core reason is that the specific trajectories used or motions performed in such works cannot guarantee the dynamical feasibility.

B. Motion Planning for MARTS

Several works [30], [31], [32], [33] have investigated collaborative manipulation and transportation of a payload by the MARTS based on the quasi-static assumption. However, these methods are severely limited because the inertial forces in agile transportation cannot be ignored simply. Manubens et al. [34] use the Transition-based RRT to search for a collision-free nonsmooth path for the MARTS. However, only simple dynamics are taken into account, so it is equally unsuitable for agile transportation. Geng et al. [35] construct a trajectory optimization problem with simplified dynamical feasibility constraints and a simplified objective function to generate trajectories for the MARTS in free space. Jackson et al. [36] simplify the obstacle as a cylinder and further parallelize the solving of the optimization problem to mitigate the explosion of optimization time as the number of aerial robots increases. Nevertheless, it still takes a few seconds to converge, even when the MARTS consists of only three aerial robots. Besides, the quality of the optimized trajectory is questionable since they only enforce an incomplete kinematic constraint (relative distance constraint) between each aerial robot and payload and neglect high-order constraints on relative velocity and acceleration. Sun and Franchi [37] consider the complete kinematic constraint and dynamical feasibility of the trajectory and propose a real-time NMPC with high scalability to the number of aerial robots. However, they also neglect the collision avoidance of the MARTS in complex environments. Wahba et al. [38] consider not only the complete kinematic constraint but also the dynamical feasibility and the collision avoidance of the MARTS in complex environments. However, they still do not address the real-time problem of trajectory planning.

Sreenath and Kumar [39] reveal that the MARTS is differentially flat when all the cables are tight and plan a smooth trajectory via minimizing the sixth derivative of flat output. However, this planning scheme neglects not only the collision

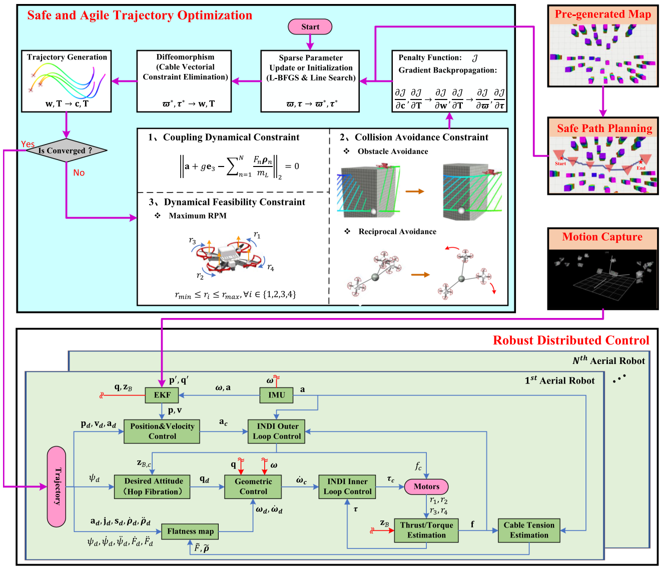


Fig. 2. Overview of our centralized trajectory planning scheme and robust distributed control scheme for the MARTS.

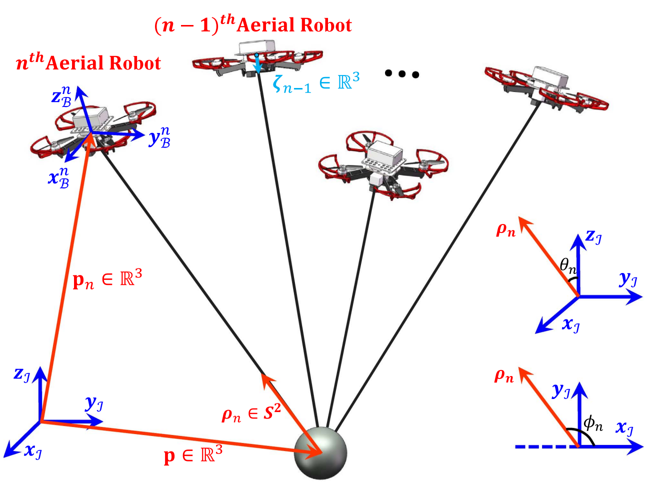


Fig. 3. Illustration of the cable-suspended payload transportation by the MARTS and the definition of θ_n and ϕ_n used to represent $\boldsymbol{\rho}_n$.

avoidance but also the dynamical feasibility of the planned trajectory. Besides, the deformation efficiency of the minimum-snap trajectory class [40], on which this scheme relies, cannot guarantee the real-time performance of trajectory planning.

III. System Overview and Preliminaries

A. System Architecture

The overall architecture of our centralized planning scheme and robust distributed control scheme is illustrated in Fig. 2. Fusing LiDAR feature points with IMU data using FAST-lio2 [41] package, we construct environmental maps. A system-level global path planning method (see Section V-A) that treats the MARTS as a convex and scalable orthopyramid is designed to find a global safe path, which acts as the initial value of sparse parameters for the back-end trajectory optimization.

For each iteration of the safe and agile trajectory optimization, by diffeomorphisms (see Section V-C1), optimization variables (auxiliary parameters) are first mapped to a set of sparse parameters required to uniquely represent the trajectory of the MARTS. Then, by solving a system of linear equations with just linear spatiotemporal computational complexity (see Section IV-B), a new set of polynomial parameters can be calculated from these sparse parameters, thereby completing the generation of a new trajectory. Next, we calculate the RPMs for each aerial robot using the flatness maps (see Section IV-A) such that the trajectory can be evaluated in terms of the collision avoidance (see Section IV-C), the dynamical feasibility (see Section IV-D), and the coupled dynamical constraint (see Section IV-E) specifically existing in our scheme. Furthermore, the total penalty function and its gradients w.r.t. polynomial parameters can be calculated sequentially. Finally, by the chain rule, these gradients are backpropagated in turn to gradients w.r.t. sparse parameters and optimization variables, based on which optimization variables update to new values (see Section V-D). The optimized trajectory will be transmitted to each aerial robot when the optimization satisfies the condition of convergence.

The odometry for each aerial robot is obtained via fusing the data from the motion capture system with the onboard IMU data by an extended Kalman filter (EKF). Our robust distributed control scheme mainly includes a two-loop control method shared by all aerial robots. The motors' RPMs are measured to estimate the thrust and control torque of the aerial robot. The outer loop adopts the incremental nonlinear dynamic inversion (INDI) [42] to get the thrust vector command based on the closed-loop trajectory tracking control law. The inner loop adopts the hop fibration rotation factorization to reconstruct the desired attitude and calculate the desired body rate as well as angular acceleration by flatness maps, which also uses INDI to get the control torque command based on the geometrical attitude control law. Finally, control commands are transformed into the signals to control each motor's RPM.

B. Dynamics Model

In this article, we consider a point-mass payload transported by a MARTS consisting of N aerial robots, as illustrated in Fig. 3. Necessary symbols to describe the dynamics model are defined in the Nomenclature. The payload is suspended on each aerial robot by a massless cable. It should be pointed out that frequent transitions of the cable between slack mode and taut mode will lead to the switching of the dynamics model, which is detrimental to control and can even cause dangerous system collapse. To address this risk, our planner will avoid the slack mode and generate a trajectory that always maintains the taut mode. Therefore, we only need to consider the following consistent dynamical model for the taut mode. The dynamics model of the payload is written as

$$\dot{\mathbf{p}} = \mathbf{v} \tag{1a}$$

$$\dot{\mathbf{v}} = -g\mathbf{e}_3 + \sum_{n=1}^{N} F_n \boldsymbol{\rho}_n / m_L. \tag{1b}$$

Besides, all the aerial robots share the same dynamics model, and the dynamics model of the nth aerial robot is written as

$$\dot{\mathbf{p}}_n = \mathbf{v}_n \tag{2a}$$

$$\dot{\mathbf{v}}_n = -g\mathbf{e}_3 + f_n \mathbf{R}_n \mathbf{e}_3 - F_n \boldsymbol{\rho}_n / m_n \tag{2b}$$

$$\dot{\mathbf{R}}_n = \mathbf{R}_n \hat{\boldsymbol{\omega}}_n \tag{2c}$$

$$\mathbf{J}_n \dot{\boldsymbol{\omega}}_n = -\boldsymbol{\omega}_n \times \mathbf{J}_n \boldsymbol{\omega}_n - F_n \boldsymbol{\zeta}_n \times \mathbf{R}_n^T \boldsymbol{\rho}_n + \boldsymbol{\tau}_n \tag{2d}$$

where $\hat{\cdot}$ is the operator to get the skew-symmetric matrix.

It should be noted that the torque $-F_n \boldsymbol{\zeta}_n \times \mathbf{R}_n^T \boldsymbol{\rho}_n$ in (2d) induced by the fact that the cable cannot always be strictly tethered to the aerial robot's center of mass (CoM) is proportional to the payload's mass m_L and the tethering error $\boldsymbol{\zeta}_n$, which is extremely important to the MARTS since it exacerbates the differences in RPMs among the aerial robot's motors. To the best of the authors' knowledge, this work is the first to incorporate this torque into the MARTS's dynamics model, which will help improve the accuracy of the planned results.

C. Kinematic Constraint Elimination

The nth taut cable imposes a kinematic constraint between the payload and the nth aerial robot

$$\|\mathbf{p}_n - \mathbf{p}\|_2 \equiv l. \tag{3}$$

It should be pointed out that the differentials of (3) also implicitly enforce constraints on $\mathbf{v}_n$ and $\mathbf{v}$ and their high-order derivatives. Therefore, if $\mathbf{p}_n$ and $\mathbf{p}$ are selected as independent optimization variables, we need to impose many equality constraints during the optimization process to ensure that these optimization variables and their high-order derivatives satisfy these kinematic constraints, which will greatly increase the complexity of solving the optimization problem. In this article, we use the payload's position $\mathbf{p}$ and the nth cable's direction vector $\boldsymbol{\rho}_n$ to indirectly represent the nth aerial robot's position

$$\mathbf{p}_n = \mathbf{p} + l\boldsymbol{\rho}_n \tag{4}$$

such that $\mathbf{p}$ and $\mathbf{p}_n$, as well as their high-order derivatives, always satisfy the complete kinematic constraint enforced by (3), which implies that the constraint is eliminated.

Let $\mathbf{x}_1, \mathbf{x}_2$, and $\mathbf{x}_3$ denote the three elements of a vector $\mathbf{x}$ in this work. In general, as in work [39], the three elements $\mathbf{F}_{n,1}, \mathbf{F}_{n,2}$, and $\mathbf{F}_{n,3}$ can be chosen as variables to uniquely represent $\mathbf{F}_n$. Thus, since $\mathbf{F}_n = F_n \boldsymbol{\rho}_n$, $\boldsymbol{\rho}_n$ can be obtained through normalizing $\mathbf{F}_n$ to solve $\mathbf{p}_n$ by $\mathbf{p} + l\mathbf{F}_n / \|\mathbf{F}_n\|_2$. However, when the norm $\|\mathbf{F}_n\|_2$ converges to zero, the unit vector $\mathbf{F}_n / \|\mathbf{F}_n\|_2$ are highly sensitive to $\mathbf{F}_{n,1}, \mathbf{F}_{n,2}$, and $\mathbf{F}_{n,3}$, and even slight changes in $\mathbf{F}_{n,1}, \mathbf{F}_{n,2}$, and $\mathbf{F}_{n,3}$ can cause dramatic changes in $\mathbf{p}_n$. To avoid this problem, we introduce two variables, i.e., the pitch angle θ_n and the azimuth angle ϕ_n of the nth cable, as depicted in Fig. 3. Thus, $\boldsymbol{\rho}_n$ can be represented as

$$\boldsymbol{\rho}_n = [\cos(\theta_n)\cos(\phi_n), \cos(\theta_n)\sin(\phi_n), \sin(\theta_n)]^T. \tag{5}$$

Let $\boldsymbol{\xi}_n = (\theta_n, \phi_n, F_n)^T \in \mathbb{R}^3$; thus, $F_n \boldsymbol{\rho}_n$ can be uniquely represented by $\boldsymbol{\xi}_n$. Using $\boldsymbol{\xi}_n$ as the variable also brings an advantage that the constraints related to the direction and magnitude of $F_n \boldsymbol{\rho}_n$ can be enforced on $\boldsymbol{\xi}_n$ directly without constructing nonlinear maps to calculate $\boldsymbol{\xi}_n$ from $F_n \boldsymbol{\rho}_n$ such that these constraints can be eliminated via designing a diffeomorphism conveniently (see Section V-C1).

D. Extended Flat-Output Variable

The MARTS is a differentially flat system. For the nth aerial robot, $\mathbf{p}, \boldsymbol{\xi}_n$, and ψ_n act as the flat-output variable. Compositing $\mathbf{p}$ and $\boldsymbol{\xi}_n, \psi_n$ for all $n \in [1, \ldots, N]$ yields a variable

$$\mathbf{Z} = (\mathbf{p}^T, \boldsymbol{\xi}_1^T, \psi_1, \ldots, \boldsymbol{\xi}_N^T, \psi_N)^T \in \mathbb{R}^{4N+3} \tag{6}$$

called as the extended flat-output variable for MARTS. Actually, $\mathbf{Z} \backslash \{\boldsymbol{\xi}_N\} \in \mathbb{R}^{4N}$ can act as one selection of flat-output variable for MARTS since $\mathbf{Z} \backslash \{\boldsymbol{\xi}_N\}$ is sufficient to uniformly determine $\boldsymbol{\xi}_N$, as elaborated in [39]. To avoid the complex map from $\mathbf{Z} \backslash \{\boldsymbol{\xi}_N\}$ onto $\boldsymbol{\xi}_N$, we relieve $\boldsymbol{\xi}_N$'s binding on $\mathbf{Z} \backslash \{\boldsymbol{\xi}_N\}$ and append redundant degree of freedom (DOF) $\boldsymbol{\xi}_N$ into $\mathbf{Z}$. As a result, we need to enforce a supererogatory dynamical constraint [see (1b)] onto $\mathbf{Z}$ during the optimization (see Section IV-E). The differentially flat characteristics help us optimize a trajectory just on the low-dimensional extended flat-output space $\mathbf{Z}$. For any $s \in \{s | s \in N_+, s \geq 1\}$, we denote $\mathbf{Z}^{[s-1]} \in \mathbb{R}^{s \times (4N+3)}$ as

$$\mathbf{Z}^{[s-1]} = \left(\mathbf{Z}^T, \dot{\mathbf{Z}}^T, \ldots, \mathbf{Z}^{(s-1)T}\right)^T. \tag{7}$$

IV. Planning for SAAT

Due to the difficulty of embedding the complete dynamical model of MARTS into the planning process using existing sample-based planners [34], the resulting planning paths are often nonsmooth, making it challenging to support agile transportation of MARTS. For optimization-based planning methods [35], [36], [38], trajectory optimization problems are typically discretized into nonlinear programming problems (NLPs) with a large number of optimization variables and equality constraints and then solved using high-performance general-purpose solvers. However, due to the computational efficiency limitations of general-purpose solvers, it is often difficult to support real-time planning of high-quality trajectories. Besides, the number of optimization variables and constraints required for collision avoidance and dynamical feasibility of the MARTS is approximately proportional to the number of aerial robots contained in the MARTS, which further limits the solution efficiency of general-purpose solvers.

Since flat-based methods can represent flat-output trajectories with the help of splines to avoid the enormous number of variables and equality constraints introduced due to the fine discretization of dynamics. The responsiveness of multiple planners [43], [44] for the autonomous navigation of a single aerial robot is improved with the help of the fast deformation capability of the spline. Therefore, this work will investigate the flat-based trajectory planning scheme for the MARTS to enhance the optimization efficiency of its trajectories.

In this section, we first derive the flatness maps (see Section IV-A) for solving RPMs of aerial robots using extended flat-output variables, supporting the construction of dynamical feasibility constraints. Then, we introduce a trajectory class (see Section IV-B) required to characterize the extended flat-output variables, enabling rapid deformation of trajectories. Next, we construct complete collision avoidance constraints (see Section IV-C) and dynamical feasibility constraints (see Section IV-D) to support the SAAT in complex environments. Furthermore, we establish a coupling dynamic constraint (see Section IV-E) to ensure that the trajectory of the extended flat output variable satisfies the dynamical equations. Finally, we establish cable's vectorial constraints (see Section IV-F) to mitigate the local optimality issue caused by the fact that flat-output variables satisfying the force balance condition always have multiple solutions.

A. Flatness Maps

Flatness maps rely only on the flat-output variable. They can help in obtaining the state and the control input of a system and assessing the dynamical feasibility of its trajectory, while avoiding the integration of dynamical equations. For each aerial robot contained in the MARTS, in addition to the necessary thrust and torque required to perform its own maneuvers, additional thrust and torque are also needed to counteract the effects induced by the cable's tension. Due to the limited tolerance of the aerial robot's actual power battery for its RPMs, the RPMs required to generate these necessary thrust and torque are the key factor in determining whether the trajectory is dynamically feasible.

Therefore, we first derive flatness maps for the square of the nth aerial robot's RPMs based on the indirect representation [see (4)]. From (2b), we can get the mass-normalized thrust vector of the nth aerial robot

$$\mathbf{f}_n = \ddot{\mathbf{p}} + l\ddot{\boldsymbol{\rho}}_n + g\mathbf{e}_3 + F_n\boldsymbol{\rho}_n/m_n. \tag{8}$$

From (2d), we can get the control torque vector

$$\boldsymbol{\tau}_n = \mathbf{J}_n\dot{\boldsymbol{\omega}}_n + \boldsymbol{\omega}_n \times \mathbf{J}_n\boldsymbol{\omega}_n + F_n\boldsymbol{\zeta}_n \times \mathbf{R}_n^T\boldsymbol{\rho}_n. \tag{9}$$

Let $\mathbf{r}_n^{\odot 2}$ denote the Hadamard product of two $\mathbf{r}_n$. Then, the flatness map for $\mathbf{r}_n^{\odot 2}$ can be derived as

$$\mathbf{r}_n^{\odot 2} = \mathbf{M}^{-1}\begin{bmatrix} m_n f_n \\ \boldsymbol{\tau}_n \end{bmatrix}. \tag{10}$$

Finally, based on the rotation factorization known as Hopf fibration [45], we use $\mathbf{f}_n$ to reconstruct the quaternion $\mathbf{q}_n$ and further derive $\boldsymbol{\omega}_n$ and $\dot{\boldsymbol{\omega}}_n$ required to calculate $\boldsymbol{\tau}_n$, the details of which are expanded in the Appendix.

It is important to note that due to the extreme complexity, the works [1] and [39], which prove that the MARTS is differentially flat, do not explicitly derive flatness maps for the high-order states and control input of the aerial robot in the MARTS and only artificially generate rather than optimize specific trajectories. As a result, fine-grained dynamical constraints cannot be constructed based on these flatness maps to ensure that the trajectories are dynamically executable. To the best of the authors' knowledge, this work is the first to explicitly derive the flatness maps down to the motors' RPMs of the aerial robot. Moreover, to improve the accuracy of the flatness maps for these RPMs, the inevitable torque caused by the fact that the cable can never be strictly tethered to the aerial robot's CoM is taken into account.

B. Trajectory Representation

The parameterization method and the deformation efficiency of the trajectory determine the efficiency of trajectory optimization. This work adopts a state-of-the-art (SOTA) polynomial trajectory class $\mathcal{C}_{\mathbf{MINCO}}$ [46], which, compared to the minimum-snap trajectory class [40], the Bézier curve [47], and the B-spline [44], has more flexible parameters. $\mathcal{C}_{\mathbf{MINCO}}$ can help to improve the optimization efficiency of trajectories through its fast spatiotemporal deformation capability with just linear computational complexity.

First, we use a multidimensional piecewise polynomial to represent the flat-output trajectory so that the motion planning for the MARTS can be performed by the deformation of the flat-output trajectory. Let $\mathbf{Z}(t)$ denote a $(4N+3)$-dimensional M-piece polynomial of $2s$ order. The mth piece of $\mathbf{Z}(t)$ is defined as

$$\mathbf{Z}|_m(t) = \mathbf{c}_m^T\boldsymbol{\beta}(t) \;\; \forall t \in [0, T_m] \tag{11}$$

where $\mathbf{c}_m \in \mathbb{R}^{2s\times(4N+3)}$ is the coefficient matrix, $\boldsymbol{\beta}(t) = [1, t^1, \ldots, t^{2s-1}]^T$ is the natural basis, and $T_m \in \mathbb{R}_{>0}$ is the time duration of the mth piece. Thus, the flat-output trajectory $\mathbf{Z}(t)$ can be fully represented by the coefficient matrix $\mathbf{c} = (\mathbf{c}_1^T, \ldots, \mathbf{c}_M^T)^T \in \mathbb{R}^{2sM\times(4N+3)}$ and the time vector $\mathbf{T} = (T_1, \ldots, T_M)^T \in \mathbb{R}_{>0}^M$.

$\mathcal{C}_{\mathbf{MINCO}}$ provides a set of sparse parameters $\{\mathbf{w}, \mathbf{T}\}$, where $\mathbf{w} = (\mathbf{w}_1, \ldots, \mathbf{w}_{M-1})^T \in \mathbb{R}^{(4N+3)\times(M-1)}$ and for any $m' \in [1, \ldots, M-1]^T$, $\mathbf{w}_{m'} = (\mathbf{p}^{m'T}, \boldsymbol{\xi}_1^{m'T}, \psi_1^{m'}, \ldots, \boldsymbol{\xi}_N^{m'T}, \psi_N^{m'})^T \in \mathbb{R}^{4N+3}$ is the m'th junction to connect the m'th piece and the $(m'+1)$th piece. Using $\{\mathbf{w}, \mathbf{T}\}$ to determine the continuity conditions at the junctions, with the boundary conditions at the initial and final moments, $\mathcal{C}_{\mathbf{MINCO}}$ constructs a system of linear equations $\mathcal{L}$ to solve polynomial parameters $\{\mathbf{c}, \mathbf{T}\}$ from $\{\mathbf{w}, \mathbf{T}\}$

$$(\mathbf{c}, \mathbf{T}) = \mathcal{L}(\mathbf{w}, \mathbf{T}) \tag{12}$$

such that a piecewise polynomial satisfies all these conditions with minimum control effort, i.e., the flat-output trajectory $\mathbf{Z}(t)$ can be uniquely generated by solving $\mathcal{L}$ directly, while avoiding time-consuming iterative optimization.

C. Collision Avoidance Constraints

The collapse of the MARTS occurs when any part of the system collides with the obstacles in a complex environment, or when any two aerial robots within the system reciprocally collide with each other. In this section, we construct constraints to avoid obstacle collisions and reciprocal collisions between the aerial robots, and design differential metrics to quantitatively assess the extent of collision. Since $\mathbf{p}$ and $\mathbf{r}_n^{\odot 2}$ used to construct metrics in this subsection and the next subsection are solved respectively by (4) and (10) algebraically from the flat output variable $\mathbf{Z}$, which, in turn, is a function of the polynomial parameters $\mathbf{c}$ and $\mathbf{T}$, we can further derive the analytic gradient of these metrics w.r.t. $\mathbf{c}$ and $\mathbf{T}$. In the following sections, we assume that all the symbols defined in the Nomenclature are associated with a special point on the mth piece of the trajectory. The details are given as follows.

1) Obstacle Avoidance Constraint: The obstacle avoidance constraint is designed to avoid obstacle collisions. Complete obstacle avoidance means that the N surfaces swept by all cables' trajectories, the N aerial robots' trajectories, and the payload's trajectory should not be intersected by any obstacle. In this work, we maintain a Euclidean signed distance field (ESDF). To evaluate the safety margin between a point $\mathbf{p}^* \in \mathbb{R}^3$ and the obstacle, we design a differential metric $\mathcal{M}_{oa} : \mathbb{R}^3 \times \mathbb{R}_{>0} \to \mathbb{R}$ as

$$\mathcal{M}_{oa}(\mathbf{p}^*, d) = d - \mathcal{E}\left(\mathbf{p}^*\right) \tag{13}$$

where $\mathcal{E}(\mathbf{p}^*)$ is the distance between $\mathbf{p}^*$ and its closest obstacle evaluated by the ESDF and $d \in \mathbb{R}_{>0}$ is the safety distance away from obstacle.

Let $d^c_{oa} \in \mathbb{R}_{>0}$, $d^Q_{oa} \in \mathbb{R}_{>0}$, and $d^L_{oa} \in \mathbb{R}_{>0}$ denote the safety distances of the cable's sampled point, the aerial robot, and the payload, respectively. We enforce the obstacle avoidance constraint on the payload

$$\mathcal{M}_{oa}(\mathbf{p}, d^L_{oa}) < 0. \tag{14}$$

For any $n \in [1, \ldots, N]$, we enforce obstacle avoidance constraint on the nth aerial robot

$$\mathcal{M}_{oa}(\mathbf{p}_n, d^Q_{oa}) < 0. \tag{15}$$

For the nth cable, we uniformly sample K points along the cable. For each $k \in [1, \ldots, K]$ and each $n \in [1, \ldots, N]$, we enforce the obstacle avoidance constraint on the kth sampled points $\mathbf{p}^k_n$ along the nth cable

$$\mathcal{M}_{oa}(\mathbf{p}^k_n, d^c_{oa}) < 0 \tag{16}$$

where $\mathbf{p}^k_n$ can be computed as

$$\mathbf{p}^k_n = \mathbf{p} + \frac{lk}{K+1}\boldsymbol{\rho_n}. \tag{17}$$

Then, the penalty function $\mathcal{J}^0_{oa}$ for the constraint enforced on the payload in (14) is defined as

$$\mathcal{J}^0_{oa} = \mathcal{L}_\mu\left[\mathcal{M}_{oa}(\mathbf{p}, d^L_{oa})\right]. \tag{18}$$

Besides, the penalty function $\mathcal{J}^n_{oa}$ for the constraints w.r.t. both the nth aerial robot and nth cable in (15) and (16) is defined as

$$\mathcal{J}^n_{oa} = \sum_{k=1}^{K} \mathcal{L}_\mu\left[\mathcal{M}_{oa}(\mathbf{p}^k_n, d^c_{oa})\right] + \mathcal{L}_\mu\left[\mathcal{M}_{oa}(\mathbf{p}_n, d^Q_{oa})\right] \tag{19}$$

where $\mathcal{L}_\mu : \mathbb{R} \to \mathbb{R}$ is a C^2-smoothing function defined as

$$\mathcal{L}_\mu(x) = \begin{cases} 0, & x \le 0 \\ (\mu - x/2)(x/\mu)^3, & 0 < x \le \mu \\ x - \mu/2, & x > \mu \end{cases} \tag{20}$$

which smooths the transition of a truncated linear function at $x = 0$.

Finally, the gradients of $\mathcal{M}_{oa}(\mathbf{p}, d^L_{oa})$ w.r.t. $\mathbf{c}_m$ and T_m are derived as

$$\frac{\partial \mathcal{M}_{oa}(\mathbf{p}, d^L_{oa})}{\partial \boldsymbol{c}_{m,\mathbf{p}}} = -\nabla\mathcal{E}(\mathbf{p})^T \otimes \boldsymbol{\beta} \tag{21a}$$

$$\frac{\partial \mathcal{M}_{oa}(\mathbf{p}, d^L_{oa})}{\partial T_m} = -\nabla\mathcal{E}(\mathbf{p})^T \mathbf{v} \tag{21b}$$

where $\otimes$ is the Kronecker product and $\nabla\mathcal{E} : \mathbb{R}^3 \to \mathbb{R}^3$ is an operator to query the ESDF gradient.

Let $\boldsymbol{\Phi}_n$ denote $(\theta_n, \phi_n)^T$ and $\varrho_k = lk/(K+1)$. The gradients of $\mathcal{M}_{oa}(\mathbf{p}^k_n, d^c_{oa})$ w.r.t. $\mathbf{c}_m$ and T_m are computed as

$$\frac{\partial \mathcal{M}_{oa}(\mathbf{p}^k_n, d^c_{oa})}{\partial \mathbf{c}_{m,\mathbf{p}}} = -\nabla\mathcal{E}(\mathbf{p}^k_n)^T \otimes \boldsymbol{\beta} \tag{22a}$$

$$\frac{\partial \mathcal{M}_{oa}(\mathbf{p}^k_n, d^c_{oa})}{\partial \mathbf{c}_{m,\boldsymbol{\Phi}_n}} = -\varrho_k\left(\nabla\mathcal{E}(\mathbf{p}^k_n)^T \frac{\partial \boldsymbol{\rho}_n}{\partial \boldsymbol{\Phi}_n}\right) \otimes \boldsymbol{\beta} \tag{22b}$$

$$\frac{\partial \mathcal{M}_{oa}(\mathbf{p}^k_n, d^c_{oa})}{\partial \mathbf{c}_{m,F_n}} = \mathbf{0} \tag{22c}$$

$$\frac{\partial \mathcal{M}_{oa}(\mathbf{p}^k_n, d^c_{oa})}{\partial T_m} = -\left(\dot{\mathbf{p}} + \varrho_k \frac{\partial \boldsymbol{\rho}_n}{\partial \boldsymbol{\Phi}_n}\dot{\boldsymbol{\Phi}}_n\right)^T \nabla\mathcal{E}(\mathbf{p}^k_n). \tag{22d}$$

Note that $\mathbf{c}_{m,\mathbf{p}}$ denotes the submatrix of $\mathbf{c}_m$ w.r.t. the variable $\mathbf{p}$ in $\mathbf{Z}$. The gradients of $\mathcal{M}_{oa}(\mathbf{p}_n, d^Q_{oa})$ w.r.t. $\mathbf{c}_m$ and T_m can be computed similarly as in (22a)–(22d) except that $\mathbf{p}^k_n$, d^c_{oa}, and ϱ_k need to be substituted by $\mathbf{p}_n$, d^Q_{oa}, and 1, which are omitted here.

2) Reciprocal Avoidance Constraint: The reciprocal avoidance constraint is designed to avoid reciprocal collisions among the aerial robots. To evaluate the safety margin between two aerial robots, we design a differential metric $\mathcal{M}_{ra} : \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}_{>0} \to \mathbb{R}$ as

$$\mathcal{M}_{ra}(\mathbf{p}, \mathbf{p}', d) = d^2 - \left\|\mathbf{p} - \mathbf{p}'\right\|^2. \tag{23}$$

For each $n \in [1, \ldots, N-1]$ and each $n' \in [n+1, \ldots, N]$, we enforce the reciprocal avoidance constraint between the nth aerial robot and the n'th aerial robot

$$\mathcal{M}_{ra}(\mathbf{p}_n, \mathbf{p}_{n'}, d^Q_{ra}) < 0 \tag{24a}$$

where $d^Q_{ra} \in \mathbb{R}_{>0}$ is the safety distance between the aerial robots. Then, the penalty function $\mathcal{J}_{ra}$ for all the reciprocal avoidance constraints is defined as

$$\mathcal{J}_{ra} = \sum_{i=1}^{N-1}\sum_{j=i+1}^{N} \mathcal{L}_\mu\left[\mathcal{M}_{ra}(\mathbf{p}_i, \mathbf{p}_j, d^Q_{ra})\right]. \tag{25}$$

Finally, the gradients of $\mathcal{M}_{ra}(\mathbf{p}_i, \mathbf{p}_j, d^Q_{ra})$ w.r.t. $\mathbf{c}_m$ and T_m are computed as

$$\frac{\partial \mathcal{M}_{ra}(\mathbf{p}_i, \mathbf{p}_j, d^Q_{ra})}{\partial \mathbf{c}_{m,\mathbf{p}}} = \mathbf{0} \tag{26a}$$

$$\frac{\partial \mathcal{M}_{ra}(\mathbf{p}_i, \mathbf{p}_j, d^Q_{ra})}{\partial \mathbf{c}_{m,\boldsymbol{\rho}_i}} = -2l\left[(\mathbf{p}_i - \mathbf{p}_j)^T \frac{\partial \boldsymbol{\rho}_i}{\partial \boldsymbol{\Phi}_i}\right] \otimes \boldsymbol{\beta} \tag{26b}$$

$$\frac{\partial \mathcal{M}_{ra}(\mathbf{p}_i, \mathbf{p}_j, d^Q_{ra})}{\partial \mathbf{c}_{m,\boldsymbol{\rho}_j}} = 2l\left[(\mathbf{p}_i - \mathbf{p}_j)^T \frac{\partial \boldsymbol{\rho}_j}{\partial \boldsymbol{\Phi}_j}\right] \otimes \boldsymbol{\beta} \tag{26c}$$

$$\frac{\partial \mathcal{M}_{ra}(\mathbf{p}_i, \mathbf{p}_j, d_{ra}^Q)}{\partial \mathbf{c}_{m,F_i}} = \frac{\partial \mathcal{M}_{ra}(\mathbf{p}_i, \mathbf{p}_j, d_{ra}^Q)}{\partial \mathbf{c}_{m,F_j}} = \mathbf{0} \tag{26d}$$

$$\frac{\partial \mathcal{M}_{ra}(\mathbf{p}_i, \mathbf{p}_j, d_{ra}^Q)}{\partial T_m} = -2l\left(\frac{\partial \boldsymbol{\rho}_i}{\partial \boldsymbol{\Phi}_i}\dot{\boldsymbol{\Phi}}_i - \frac{\partial \boldsymbol{\rho}_j}{\partial \boldsymbol{\Phi}_j}\dot{\boldsymbol{\Phi}}_j\right)^T \cdot (\mathbf{p}_i - \mathbf{p}_j). \tag{26e}$$

D. Dynamical Feasibility Constraints

For any aerial robot in the MARTS, if the peak RPM of any motor exceeds the maximum admissible RPM of the power battery, it will cause the MARTS to crash. In this section, we enforce dynamical feasibility constraints on each aerial robot to limit its transient RPMs along the trajectory when the agility enhances as the trajectory's duration decreases, such that the optimized agile trajectory can be executed by the practical MARTS. Compared to the relatively conservative combination of constraints constructed in the literature, consisting of maximum attitude angle, maximum total thrust, and maximum angular velocity, the maximum RPM constraint designed in this work can maximize the feasible trajectory space for the MARTS. The details are given as follows.

For each $n \in [1, \ldots, N]$, we restrict the upper bound of all the nth aerial robot's RPMs

$$0 \leq r_{n,i}^2 \leq r_{\max}^2 \quad \forall i \in \{1, 2, 3, 4\} \tag{27}$$

where $r_{\max}$ is the maximum admissible RPM. Note that in (27), we add a lower bound constraint in addition to the upper bound constraint. This is because $\mathbf{r}_n^{\odot 2}$ is obtained by algebraically solving the flatness map [see (10)], so it cannot be guaranteed to be strictly nonnegative. For a scalar a, we define a metric $\mathcal{M} : \mathbb{R} \times \mathbb{R}_{>0} \to \mathbb{R}$ to evaluate whether a exceeds its bound $\bar{a}$

$$\mathcal{M}(a, \bar{a}) = a - \bar{a}. \tag{28}$$

Then, the penalty function for the constraint in (27) is defined as

$$\mathcal{J}_r^n = \sum_{i=1}^{4} \mathcal{L}_\mu \left(\mathcal{M}_{df}\left(\left(r_{n,i}^2 - \frac{r_{\max}^2}{2}\right)^2, \frac{r_{\max}^4}{4}\right)\right). \tag{29}$$

And the gradients of $\mathcal{M}_{df}$ w.r.t. $\mathbf{c}_m$ and T_m are computed as

$$\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{c}_{m,\mathbf{p}}} = \sum_{i\in\Pi}\left[\left(\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{f}_n^{(i)}}\right)^T \frac{\partial \mathbf{f}_n^{(i)}}{\partial \mathbf{p}^{(i+2)}}\right] \otimes \boldsymbol{\beta}^{(i+2)} \tag{30a}$$

$$\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{c}_{m,\boldsymbol{\rho}_n}} = \sum_{i\in\Pi}\sum_{j\in\Pi^i}\sum_{k=0}^{j}\left[\left(\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{f}_n^{(i)}}\right)^T \frac{\partial \mathbf{f}_n^{(i)}}{\partial \boldsymbol{\rho}_n^{(j)}}\frac{\partial \boldsymbol{\rho}_n^{(j)}}{\partial \boldsymbol{\Phi}_n^{(k)}}\right] \otimes \boldsymbol{\beta}^{(k)} + \left[\left(\mathbf{M}^{-T}\frac{\partial \mathcal{M}_{df}}{\partial \boldsymbol{r}_n^{\odot 2}}\right)_{[2:4]}^T \frac{\partial \boldsymbol{\tau}_n}{\partial \boldsymbol{\rho}_n}\frac{\partial \boldsymbol{\rho}_n}{\partial \boldsymbol{\Phi}_n}\right] \otimes \boldsymbol{\beta} \tag{30b}$$

$$\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{c}_{m,F_n}} = \sum_{i\in\Pi}\sum_{j=0}^{i}\left[\left(\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{f}_n^{(i)}}\right)^T \frac{\partial \mathbf{f}_n^{(i)}}{\partial F_n^{(j)}}\right]\boldsymbol{\beta}^{(j)} + \left[\left(\mathbf{M}^{-T}\frac{\partial \mathcal{M}_{df}}{\partial \boldsymbol{r}_n^{\odot 2}}\right)_{[2:4]}^T \frac{\partial \boldsymbol{\tau}_n}{\partial F_n}\right]\boldsymbol{\beta} \tag{30c}$$

$$\begin{aligned}\frac{\partial \mathcal{M}_{df}}{\partial T_m} = &\sum_{i\in\Pi}\sum_{j\in\Pi^i}\sum_{k=0}^{j}\left(\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{f}_n^{(i)}}\right)^T \frac{\partial \mathbf{f}_n^{(i)}}{\partial \boldsymbol{\rho}_n^{(j)}}\frac{\partial \boldsymbol{\rho}_n^{(j)}}{\partial \boldsymbol{\Phi}_n^{(k)}}\boldsymbol{\Phi}_n^{(k+1)} \\ &+ \left(\mathbf{M}^{-T}\frac{\partial \mathcal{M}_{df}}{\partial \boldsymbol{r}_n^{\odot 2}}\right)_{[2:4]}^T \frac{\partial \boldsymbol{\tau}_n}{\partial \boldsymbol{\rho}_n}\frac{\partial \boldsymbol{\rho}_n}{\partial \boldsymbol{\Phi}_n}\dot{\boldsymbol{\Phi}}_n \\ &+ \sum_{i\in\Pi}\sum_{j=0}^{i}\left(\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{f}_n^{(i)}}\right)^T \frac{\partial \mathbf{f}_n^{(i)}}{\partial F_n^{(j)}}F_n^{(j+1)} \\ &+ \left(\mathbf{M}^{-T}\frac{\partial \mathcal{M}_{df}}{\partial \boldsymbol{r}_n^{\odot 2}}\right)_{[2:4]}^T \frac{\partial \boldsymbol{\tau}_n}{\partial F_n}\dot{F}_n \\ &+ \sum_{i\in\Pi}\left(\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{f}_n^{(i)}}\right)^T \frac{\partial \mathbf{f}_n^{(i)}}{\partial \mathbf{p}^{(i+2)}}\mathbf{p}^{(i+3)}\end{aligned} \tag{30d}$$

where the subscript [1] denotes the first element of the 4-D vector, and $[2:4]$ denotes the 3-D vector composed of the second, third, and fourth elements of this vector, and the index sets are defined, respectively, as $\Pi = \{0, 1, 2\}$, $\Pi^0 = \{0, 2\}$, $\Pi^1 = \{0, 1, 3\}$, and $\Pi^2 = \{0, 1, 2, 4\}$. Besides, for any $i \in \Pi$, $\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{f}_n^{(i)}}$ can be further calculated as

$$\begin{aligned}\frac{\partial \mathcal{M}_{df}}{\partial \mathbf{f}_n^{(i)}} = &\left(\mathbf{M}^{-T}\frac{\partial \mathcal{M}_{df}}{\partial \boldsymbol{r}_n^{\odot 2}}\right)_{[2:4]}^T \left(\sum_{i}^{0}\sum_{j,k}^{3}\frac{\partial \boldsymbol{\tau}_n}{\partial (\mathbf{R}_n)_{j,k}}\frac{\partial (\mathbf{R}_n)_{j,k}}{\partial \mathbf{f}_n^{(i)}}\right. \\ &\left. + \sum_{j=i}^{1}\frac{\partial \boldsymbol{\tau}_n}{\partial \boldsymbol{\omega}_n}\frac{\partial \boldsymbol{\omega}_n}{\partial \mathbf{z}_{\mathcal{B}}^{n(j)}}\frac{\mathbf{z}_{\mathcal{B}}^{n(j)}}{\partial \mathbf{f}_n^{(i)}} + \sum_{j=i}^{2}\frac{\partial \boldsymbol{\tau}_n}{\partial \dot{\boldsymbol{\omega}}_n}\frac{\partial \dot{\boldsymbol{\omega}}_n}{\partial \mathbf{z}_{\mathcal{B}}^{n(j)}}\frac{\mathbf{z}_{\mathcal{B}}^{n(j)}}{\partial \mathbf{f}_n^{(i)}}\right) \\ &+ m_n\left(\mathbf{M}^{-T}\frac{\partial \mathcal{M}_{df}}{\partial \boldsymbol{r}_n^{\odot 2}}\right)_{[1]}^T \sum_{i}^{0}\frac{\partial f_n}{\partial \mathbf{f}_n^{(i)}}.\end{aligned} \tag{31}$$

E. Coupling Dynamic Constraint

As exposited in Section III-D, we need to enforce an extra payload dynamical constraint on the extended flat-output variable $\mathbf{Z}$ based on (1b)

$$\dot{\mathbf{v}} \equiv -g\mathbf{e}_3 + \sum_{n=1}^{N} F_n \boldsymbol{\rho}_n / m_L. \tag{32}$$

Then, we construct the penalty function for this constraint

$$\mathcal{J}_d = \mathcal{M}\left(\left\|\ddot{\mathbf{p}} + g\mathbf{e}_3 - \sum_{n=1}^{N} F_n \boldsymbol{\rho}_n / m_L\right\|, 0\right). \tag{33}$$

Denote by $\mathcal{M}_{cd}$ the right-hand side of (32), and the gradients of $\mathcal{M}_{cd}$ w.r.t. $\mathbf{c}_m$ and T_m are computed as

$$\frac{\partial \mathcal{M}_{cd}}{\partial \mathbf{c}_{\mathbf{p}}} = \left(\frac{\partial \mathcal{M}_{cd}}{\partial \ddot{\mathbf{p}}}\right)^T \otimes \ddot{\boldsymbol{\beta}} \tag{34a}$$

$$\frac{\partial \mathcal{M}_{cd}}{\partial \mathbf{c}_{m,\boldsymbol{\rho}_n}} = \left[\left(\frac{\partial \mathcal{M}_{cd}}{\partial \boldsymbol{\rho}_n}\right)^T \frac{\partial \boldsymbol{\rho}_n}{\partial \boldsymbol{\Phi}_n}\right] \otimes \boldsymbol{\beta} \tag{34b}$$

$$\frac{\partial \mathcal{M}_{cd}}{\partial \mathbf{c}_{m,F_n}} = \frac{\partial \mathcal{M}_{cd}}{\partial F_n}\boldsymbol{\beta} \tag{34c}$$

$$\frac{\partial \mathcal{M}_{cd}}{\partial T_m} = \sum_{n=1}^{N} \left[\left(\frac{\partial \mathcal{M}_{cd}}{\partial \boldsymbol{\rho}_n} \right)^T \frac{\partial \boldsymbol{\rho}_n}{\partial \boldsymbol{\Phi}_n} \dot{\boldsymbol{\Phi}}_n + \frac{\partial \mathcal{M}_{cd}}{\partial F_n} \dot{F}_n \right] + \left(\frac{\partial \mathcal{M}_{cd}}{\partial \ddot{\mathbf{p}}} \right)^T \dddot{\mathbf{p}}. \tag{34d}$$

F. Cable's Vectorial Constraints

Considering an acceleration $\mathbf{a}$ and a set $\mathcal{F} = \{F_1\boldsymbol{\rho}_1, \ldots, F_N\boldsymbol{\rho}_N\}$ that contains all cables' force vectors satisfying (1b), we find that $\mathbf{a}$ and a new set $\mathcal{F}' = \{R(\phi)F_1\boldsymbol{\rho}_1, \ldots, R(\phi)F_N\boldsymbol{\rho}_N\}$ obtained by rotating all the vectors in $\mathcal{F}$ around the vector $\mathbf{a} + g\mathbf{e}_3$ by an arbitrary angle ϕ still satisfy (1b), where $R(\phi) \in SO(3)$ is the rotation matrix. Besides, from (1b), we also conclude that the set $\mathcal{F}'' = \{F_{\sigma(1)}\boldsymbol{\rho}_{\sigma(1)}, \ldots, F_{\sigma(N)}\boldsymbol{\rho}_{\sigma(N)}\}$ obtained by commutating the force vectors among the cables does not vary the payload's dynamics, where $\sigma : \mathcal{Z}^+ \to \mathcal{Z}^+$ is the index commutation operator.

Then, let us analyze the reasons for local optima that are prone to occur in the trajectory planning of the MARTS. At the start of trajectory optimization, the coupling dynamic constraints [see (32)] are not satisfied. As optimization proceeds, each local trajectory segment undergoes independent deformation to reduce violations of the coupling dynamic constraint and minimize its local trajectory energy. However, the reduction in violations of the coupling dynamic constraint and the decrease in local trajectory energy are not always synchronized. During optimization, a local trajectory segment may undergo severe deformation to quickly satisfy the coupling dynamic constraint, but severe deformation means an increase in the local trajectory energy. If the combination of cables' force vectors within this local trajectory segment differs from those of its preceding and subsequent local trajectory segments, then, to further reduce the overall trajectory energy, this local trajectory segment must readjust its combination of cables' force vectors to align with those of its preceding and subsequent local segments. However, due to the premature convergence of the coupling dynamic constraint, it becomes difficult to violate the coupling dynamic constraint again to readjust the combination of cables' force vectors, leading to a locally energy-optimal trajectory. Besides, infinite feasible sets like $\mathcal{F}'$ and $\mathcal{F}''$ further increase the possibility of this local optimal problem.

To alleviate the local optima and speed up the convergence of the optimization, we separate the vectorial range of the cables and limit the optimization of each force vector within its own vectorial range. Concretely, for each $n \in [1, \ldots, N]$, we construct the following vectorial constraints:

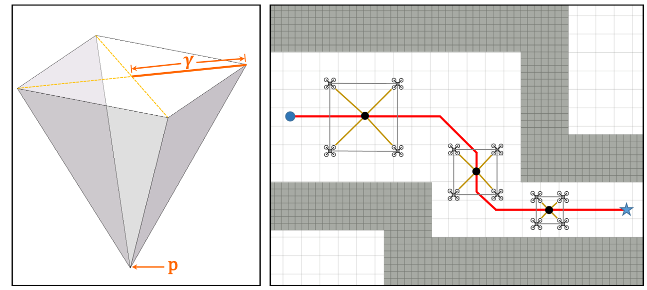
$$0 \le \theta_n \le \theta_{\max} \tag{35a}$$

$$-\frac{(2n-3)\pi}{N} \le \phi_n \le \frac{(2n-1)\pi}{N} \tag{35b}$$

where $\theta_{\max}$ is the maximum allowable pitch angle of the cable. Besides, we also enforce the constraint on each cable's force F_n

$$F_{\min} \le F_n \le F_{\max} \tag{36}$$

where $F_{\min}$ and $F_{\max}$ are, respectively, the minimum and maximum allowable forces of the cable. This constraint prevents the cable from being slack and providing an excessive force compared to other cables.



(a) (b)

Fig. 4. Illustration of the system-level path planning method proposed in Section V-A. (a) Simplified configuration of the MARTS. (b) Simplified configuration scales to fit different widths of the corridor.

V. Spatial–Temporal Trajectory Optimization

In this section, we will construct a complete planning scheme for the SAAT of cable-suspended payloads by the MARTS in complex environments. First, we design a system-level front-end path planning method (see Section V-A) for the MARTS to achieve rapid search for collision-free paths used as initial values for back-end trajectory optimization. Then, we construct the back-end trajectory optimization problem (see Section V-B) for the SAAT of the MARTS in complex environments. Besides, to simplify the trajectory optimization problem and improve its optimization efficiency, we design diffeomorphisms (see Sections V-C1 and V-C2) to eliminate the manifold constraints enforced on temporal variables of the trajectory and the geometric constraints enforced on the cable's force vector. Furthermore, we eliminate the continuous-time constraints in the optimization problem (see Section V-C3) by solving the integration of constraint violations approximatively, thereby transforming the original trajectory optimization problem into a lightweight unconstrained optimization problem (see Section V-D) that can be solved efficiently.

A. System-Level Path Planning

To provide a reasonable initial path for trajectory optimization, we design a system-level global path planning method. The MARTS is simplified to a mobile and scalable regular pyramid, and the length of its lateral edge is equal to the length of the cable. As shown in Fig. 4, γ is defined as the scalable variable used to regulate the size of the regular polygon base. Let $\mathcal{C}_{rp}$ be the configuration space of this regular pyramid. Thus, given the payload's position $\mathbf{p}$ and γ, any configuration in $\mathcal{C}_{rp}$ can be uniquely represented by an ordered pair $(\mathbf{p}, \gamma)$. We discretize $\mathcal{C}_{rp}$ and use A^* algorithm to obtain a collision-free path. A collision-free configuration on this path signifies that the entire regular pyramid with γ at point $\mathbf{p}$ does not collide with any environmental obstacle. We use a solid, convex, and even conservative regular pyramid instead of a hollow nonconvex compact geometry just enveloping the entire MARTS to check for collisions such that a safe corridor that is not penetrated by any obstacle can be opened up by uniting the consecutive convex regular pyramids for trajectory deformation in the back-end optimization.

B. Trajectory Optimization Problem Formulation

The trajectory optimization problem for the SAAT can be formulated as follows:

$$\min_{\mathbf{w},\mathbf{T}} \mathcal{J}_E = \int_0^T \|\mathbf{p}^{(s)}(t)\|^2 dt + \lambda_Z \sum_{n=1}^{N} \int_0^T \|\boldsymbol{\xi}_n^{(s)}(t)\|^2 dt + \lambda_T T_\Sigma \tag{37a}$$

$$\text{s.t. } (\mathbf{c},\mathbf{T}) = \mathcal{L}(\mathbf{w},\mathbf{T}) \tag{37b}$$

$$T_m > 0 \quad \forall m \in [1,\ldots,M] \tag{37c}$$

$$\mathbf{Z}^{[s-1]}(0) = \bar{\mathbf{Z}}_0 \tag{37d}$$

$$\mathbf{Z}^{[s-1]}(T_\Sigma) = \bar{\mathbf{Z}}_{T_\Sigma} \tag{37e}$$

$$\mathcal{G}(\mathbf{Z}^{[s-1]}(t)) \preceq \mathbf{0} \quad \forall t \in [0, T_\Sigma] \tag{37f}$$

$$\mathcal{H}(\mathbf{Z}^{[s-1]}(t)) = \mathbf{0} \quad \forall t \in [0, T_\Sigma] \tag{37g}$$

where $T_\Sigma = \sum_{m=1}^{M} T_m$. Cost function (37a) balances the smoothness and agility of the trajectory $\mathbf{Z}(t)$, λ_Z regulates the smoothness of the aerial robot's motion related to the load, and λ_T is the time regularization parameter. Equation (37b) restricts that the polynomial parameters $\{\mathbf{c},\mathbf{T}\}$ representing $\mathbf{Z}(t)$ can only be mapped by sparse parameters $\mathbf{w}$ and $\mathbf{T}$ provided by $\mathcal{C}_{\mathbf{MINCO}}$. Equation (37c) guarantees that the duration of each piece is positive. Equations (37d) and (37e) set the trajectory's boundary conditions. Equation (37f) denotes the inequality constraints continuously imposed along the trajectory, and (37g) denotes additional equality constraints designed to ensure that the flat-output trajectory $\mathbf{Z}(t)$ complies with the payload's dynamics.

Then, we derive the formulations of $\mathcal{J}_E$ gradient w.r.t. $\mathbf{c}_m$ and T_m as

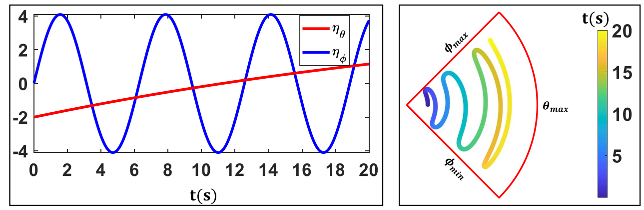
$$\frac{\partial \mathcal{J}_E}{\partial \mathbf{c}_{m,\mathbf{p}}} = 2\left(\int_0^{T_m} \boldsymbol{\beta}^{(s)}(t)\boldsymbol{\beta}^{(s)}(t)^T dt\right)\mathbf{c}_{m,\mathbf{p}} \tag{38a}$$

$$\frac{\partial \mathcal{J}_E}{\partial \mathbf{c}_{m,\boldsymbol{\xi}_n}} = 2\lambda_Z\left(\int_0^{T_m} \boldsymbol{\beta}^{(s)}(t)\boldsymbol{\beta}^{(s)}(t)^T dt\right)\mathbf{c}_{m,\boldsymbol{\xi}_n} \tag{38b}$$

$$\frac{\partial \mathcal{J}_E}{\partial T_m} = \left\|\mathbf{p}^{(s)}(T_m)\right\|^2 + \lambda_Z \sum_{n=1}^{N}\left\|\boldsymbol{\xi}_n^{(s)}(T_m)\right\|^2 + \lambda_T. \tag{38c}$$

C. Constraint Elimination

An infinite number of constraints are introduced since the inequality (37f) and the equality (37g) are enforced over the trajectory's entire duration T_Σ, leading to difficulties in solving the optimization problem (37). Therefore, to simplify this optimization problem, we design approaches to eliminate the infinite number of constraints as well as the temporal constraints (37c) as follows.



(a) (b)

Fig. 5. Illustration of the diffeomorphism defined in (39). (a) Selected trajectories for the auxiliary variables $\eta_\theta \in \mathbb{R}, \eta_\phi \in \mathbb{R}$ to solve the trajectories of θ, ϕ by (39). (b) Trajectory of $\boldsymbol{\rho}_n$ solved by (5) using the trajectories of θ and ϕ can always be compressed into a bounded red fan-shaped region.

1) Vectorial Constraint Elimination: Note that vectorial constraints [see (35) and (36)] enforced on the nth force vector are constructed directly on $\boldsymbol{\xi}_n$. For each $m' \in [1,\ldots,M-1]$, let us introduce an auxiliary variable $\boldsymbol{\eta}_n^{m'} \in \mathbb{R}^3$ for $\boldsymbol{\xi}_n^{m'}$. Then, for each element $\xi \in \boldsymbol{\xi}_n^{m'}$, we design a smooth diffeomorphism $\mathcal{S}: \mathbb{R} \to \mathbb{R}$ and select the same element $\eta \in \boldsymbol{\eta}_n^{m'}$ to map ξ as

$$\xi = \mathcal{S}(\eta) = \frac{\xi_{\min} + \xi_{\max}}{2} + \frac{(\xi_{\max} - \xi_{\min})}{\pi} \arctan \eta \tag{39}$$

where $\xi_{\min}$ and $\xi_{\max}$ are, respectively, the lower and upper bounds of ξ. $\mathcal{S}$ maps $\mathbb{R}$ to interval $[\xi_{\min}, \xi_{\max}]$ such that arbitrary optimization of η on $\mathbb{R}$ always guarantees $\xi \in [\xi_{\min}, \xi_{\max}]$. As shown in Fig. 5, as a result, the sparse parameters $\{\mathbf{w},\mathbf{T}\}$ can be substituted by the auxiliary parameters $\{\boldsymbol{\varpi},\boldsymbol{\tau}\}$ acting as the actual optimization variables, where $\boldsymbol{\varpi} = (\boldsymbol{\varpi}_1,\ldots,\boldsymbol{\varpi}_{M-1})^T \in \mathbb{R}^{(4N+3)\times(M-1)}$ and for any $m' \in [1,\ldots,M-1]$, $\boldsymbol{\varpi}_{m'} \in \mathbb{R}^{4N+3}$ can be obtained through replacing $\boldsymbol{\xi}_n^{m'}$ by $\boldsymbol{\eta}_n^{m'}$ for all $n \in [1,\ldots,N]$ and denoted by $\boldsymbol{\varpi}_{m'} = (\mathbf{p}^{m'^T}, \boldsymbol{\eta}_1^{m'^T}, \psi_1^{m'}, \ldots, \boldsymbol{\eta}_N^{m'^T}, \psi_N^{m'})^T$. The optimization variables $\{\boldsymbol{\varpi},\boldsymbol{\tau}\}$ as well as $\mathcal{S}$ help to eliminate these vectorial constraints enforced at the m'th junction $\boldsymbol{\xi}_n^{m'} \in \mathbf{w}_{m'}$ between the m'th piece and the $(m'+1)$th piece of trajectory.

Furthermore, the gradient of any penalty function $\mathcal{J}$ w.r.t. η can be computed as

$$\frac{\partial \mathcal{J}}{\partial \eta} = \frac{\xi_{\max} - \xi_{\min}}{\pi(\eta^2+1)} \frac{\partial \mathcal{J}}{\partial \xi}. \tag{40}$$

2) Temporal Constraint Elimination: In order to eliminate the constraint (37c) on T_m, we introduce $\tau = [\tau_1,\ldots,\tau_M]^T \in \mathbb{R}^M$ as an auxiliary temporal vector and define the following diffeomorphism to map T_m from τ_m:

$$T_m = e^{\tau_m} \tag{41}$$

such that optimizing τ_m over $\mathbb{R}$ always guarantees $T_m > 0$.

3) Infinite Continuous-Time Constraint Elimination: Inspired by work [48], to transform infinite inequality constraints to finite constraints, we adopt the constraint transcription by introducing a new integral-type constraint as

$$\mathcal{J}_{\mathcal{G}}^{\int} = \int_0^{T_\Sigma} \mathcal{J}_{\mathcal{G}} dt \leq 0. \tag{42}$$

Since the analytic value of $\mathcal{J}_{\mathcal{G}}^{\int}$ is always difficult to evaluate, we turn to approximating it by quadrature using uniformly sampled penalty functions along the trajectory as

$$\mathcal{J}_{\mathcal{G}}^{\int} \approx \mathcal{J}_{\mathcal{G}}^{\Sigma} = \sum_{m=1}^{M} \mathcal{J}_{\mathcal{G},m}^{\Sigma} \tag{43}$$

where $\mathcal{J}_{\mathcal{G},m}^{\Sigma}$ is the approximate quadrature for the mth piece can be calculated as

$$\mathcal{J}_{\mathcal{G},m}^{\Sigma} = \frac{T_m}{\kappa} \sum_{k=0}^{\kappa} \bar{\omega}_k \mathcal{J}_{\mathcal{G}} \left(\mathbf{Z}|_m^{[s-1]}(t_k) \right). \tag{44}$$

κ is the sampled number on each piece, $(\bar{\omega}_0, \bar{\omega}_1, \ldots, \bar{\omega}_{\kappa_i - 1}, \bar{\omega}_{\kappa_i}) = (1/2, 1, \ldots, 1, 1/2)$ are the coefficients of the sampled penalty functions for the quadrature following the trapezoidal rule [49], and $t_k = \frac{k}{\kappa} T_m$. Then, the gradients of $\mathcal{J}_{\mathcal{G},m}^{\Sigma}$ w.r.t. $\mathbf{c}_m$ and T_m can be easily derived as

$$\frac{\partial \mathcal{J}_{\mathcal{G},m}^{\Sigma}}{\partial \mathbf{c}_m} = \frac{T_m}{\kappa} \sum_{k=0}^{\kappa} \bar{\omega}_k \frac{\partial \mathcal{J}_{\mathcal{G}} \left(\mathbf{Z}|_m^{[s-1]}(t_k) \right)}{\partial \mathbf{c}_m} \tag{45a}$$

$$\frac{\partial \mathcal{J}_{\mathcal{G},m}^{\Sigma}}{\partial T_m} = \frac{\mathcal{J}_{\mathcal{G},m}^{\Sigma}}{T_m} + \frac{T_m}{\kappa^2} \sum_{k=0}^{\kappa} k \bar{\omega}_k \frac{\partial \mathcal{J}_{\mathcal{G}} \left(\mathbf{Z}|_m^{[s-1]}(t_k) \right)}{\partial t_k}. \tag{45b}$$

Since the DOF of optimization variable $\mathbf{Z}$ related to the coupling dynamic constraint [see (32)] is $3 + 3 \times N$, which is much larger than 3, i.e., the number of this constraint, we can make the trajectory approximately satisfy (32) by constructing a finite constraint $\mathcal{J}_{\mathcal{H}}^{\int} \leq 0$ similar to (43) and (44) and approximately calculate $\mathcal{J}_{\mathcal{H}}^{\Sigma}$ and $\mathcal{J}_{\mathcal{H},m}^{\Sigma}$ similar to (43) and (44). Besides, the gradients of $\mathcal{J}_{\mathcal{H},m}^{\Sigma}$ w.r.t. $\mathbf{c}_m$ and T_m are similar to (45a) and (45b), which are abandoned here.

D. Unconstrained NLP Formation

After the elimination of the vectorial constraints in Section V-C1, the continuous-time penalty functions for the SAAT in complex environments can be summarized as

$$\mathcal{J}_{\mathcal{G}} = \sum_{n=0}^{N} \lambda_{oa} \mathcal{J}_{oa}^{n} + \sum_{n=1}^{N} \lambda_{\text{rpm}} \mathcal{J}_{\text{rpm}}^{n} + \lambda_{ra} \mathcal{J}_{ra} \tag{46a}$$

$$\mathcal{J}_{\mathcal{H}} = \lambda_d \mathcal{J}_d \tag{46b}$$

where $\lambda_{oa}, \lambda_{ra}, \lambda_{\text{rpm}}$, and λ_d are preset weights for the penalty functions.

Then, based on the constraint elimination introduced in Section V-C3, we convert original optimization problem defined in (37) to an unconstrained optimization problem

$$\min_{\boldsymbol{\varpi}, \boldsymbol{\tau}} \mathcal{J}_E + \sum_{m=1}^{M} \left(\mathcal{J}_{\mathcal{G},m}^{\Sigma} + \mathcal{J}_{\mathcal{H},m}^{\Sigma} \right). \tag{47}$$

This problem can be efficiently solved by fast spatiotemporal deformation provided by $\mathcal{C}_{\mathbf{MINCO}}$. Each deformation involves the following two procedures, as shown in Fig. 2. In the forward trajectory generation, sparse parameters $\{\mathbf{w}, \mathbf{T}\}$ are first mapped from optimization variables $\{\boldsymbol{\varpi}, \boldsymbol{\tau}\}$; then, $\mathcal{C}_{\mathbf{MINCO}}$ uses the banded matrix PLU decomposition to solve (12), acquiring polynomial parameters $\{\mathbf{c}, \mathbf{T}\}$ from sparse parameters $\{\mathbf{w}, \mathbf{T}\}$. Besides, in the backward gradient propagation, through ingenious reuse of the solved PLU matrixes, $\mathcal{C}_{\mathbf{MINCO}}$ avoids the inversion of a $2sM$-dimensional matrix such that the gradients $\{\frac{\partial \mathcal{J}}{\partial \mathbf{c}}, \frac{\partial \mathcal{J}}{\partial \mathbf{T}}\}$ are first propagated to $\{\frac{\partial \mathcal{J}}{\partial \mathbf{w}}, \frac{\partial \mathcal{J}}{\partial \mathbf{T}}\}$ and further propagated to $\{\frac{\partial \mathcal{J}}{\partial \boldsymbol{\varpi}}, \frac{\partial \mathcal{J}}{\partial \boldsymbol{\tau}}\}$. Both processes rely just on linear spatiotemporal computational complexity. Thus, fast spatiotemporal deformation of trajectory will be performed via the updation of $\{\boldsymbol{\varpi}, \boldsymbol{\tau}\}$.

It should be pointed out that although our planning scheme requires the payload's state (position and its high-order derivatives) and even the cables' states to be provided as the boundary conditions [see (37d) and (37e)] for trajectory optimization, this does not imply that we need to estimate these states in real time and use them as the initial values for continuous online replannings. We first provide stable hovering states for the MARTS, in which each aerial robot is equally spaced and uniformly loaded according to force equilibrium, the force vector of each cable is predefined, and the high-order derivatives of both the payload's position and each cable's force vector are zero. Given a payload's position on the map, a stable hovering state of the MARTS can be uniquely determined by the kinematic constraint and flatness maps. At the beginning, the MARTS stabilizes in a safe hovering state on the map. For the first trajectory, our planner uses this hovering state as the initial value for the trajectory optimization. When replanning is triggered, the reference state on the trajectory 200 ms after the current execution time will be used as the initial value for the next trajectory replanning. This approach ensures strict continuity between the reference values of adjacent trajectories at connection points, as well as strict continuity in the tracking errors of each aerial robot, and avoids the real-time estimation of these states. Besides, for each replanning, the other safe hovering state on the map is used as the terminal value of the trajectory.

It is important to note that Sreenath and Kumar [39] can only generate a relatively conservative dynamically feasible trajectory for transportations in obstacle-free spaces by minimizing the snap of the flat-output trajectory rather than proposing a complete flat-based planning scheme for the SAAT. They neglect not only the collision avoidance in complex environments but also the dynamical feasibility for agile transportation. Besides, the deformation efficiency of the minimum-snap trajectory class [40], on which their scheme relies, cannot guarantee the real-time performance of trajectory planning. However, our planning scheme achieves the real-time trajectory generation for the SAAT of cable-suspended payloads by the MARTS in complex environments.

VI. Control Scheme for Agile Transportation

First, we analyze the main characteristics present in the trajectory tracking control problem of the MARTS.

1) The payload lacks direct force control and can only be indirectly controlled by necessary cable tensions provided by aerial robots. The required cable tension can only be provided by first adjusting the aerial robot's attitude and

thrust. The inherent response lag in this process fundamentally limits the tracking precision of the payload.

2) Achieving precise closed-loop control of the payload relies on accurate control of the cable's tension vector. For our experiments, deploying tension sensors and IMUs on the cable for real-time measurements of the tension vector is overly complex. Therefore, although the cable's tension vector can be estimated in real time, the precise control of the cable tension remains challenging, which further limits the tracking accuracy of the payload.
3) For practical aerial transportation, placing sensors on diverse payloads for direct state measurements is impractical. Even if the payload's state can be estimated by collecting all aerial robots' states, factors like variance in data time stamps, communication delays, noise, and estimation errors will degrade the estimation accuracy of the payload's state, introducing uncertainty into the control precision of the payload.

Considering the theoretical limitations, the challenges in engineering deployment, and the numerous potential uncertainties analyzed above, we abandon the pursuit of closed-loop control for the payload and instead permit the payload to always exhibit tracking error. In this work, we integrate a robust distributed control scheme to achieve agile transportation, the architecture of which is shown at the bottom of Fig. 2. It should be pointed out that in this section, we add n to the superscript or subscript of symbols to specify that they belong to the nth aerial robot, but for simplicity, we omit n in Fig. 2.

For our control scheme, the position and velocity errors of the payload would not be considered in the aerial robot's control law, such that the system oscillation induced by the payload's control law and the time delay induced by control signal transmission can be avoided fundamentally. INDI is used to estimate the direction and magnitude of the tension in each cable, such that we do not need to deploy additional tension sensors or attitude sensors to measure these states, and the unknown payload's mass can be online estimated. This scheme gets rid of the reliance on the state measurement for both payload and cable as in [16], [25], [26], and [27], as well as the closed-loop control for payload as in [16], [22], [23], [25], [26], and [27]. Instead, it pursues closed-loop control for each aerial robot. The design details of the outer-loop trajectory tracking controller and the inner-loop geometric attitude controller contained in this scheme can be found in [29], [42], and [45], which are not elaborated here.

Then, we give the method to estimate the payload's mass. When the payload is lifted, the MARTS will be stabilized at a predefined position. Then, the mass of the payload can be estimated by a finite sequence of each aerial robot's estimations for the tension vector of the cable attached to it as follows:

$$\widetilde{m_L} = \sum_{n=1}^{N} \sum_{w=1}^{W} \frac{\widetilde{F_n \boldsymbol{\rho}_n|_{w,3}}}{W} \tag{48}$$

where $\widetilde{F_n \boldsymbol{\rho}_n}|_{w,3}$ denotes the z component of the wth estimation for $F_n \boldsymbol{\rho}_n$ by INDI in the sequence and W denotes the number of estimations in the sequence.

Since the INDI always compensates for the actual force and torque exerted by the cable, this control scheme is robust against the deviations from the planned trajectory, which can be introduced by the payload's swing around the cables' attachment point induced by the inevitable attachment error, the estimation error of the payload's mass, the unsynchronized reception of trajectory among aerial robots due to communication delay with variance, and wind disturbances.

Although we introduce the torque caused by the cable (whose attachment point cannot be strictly at the aerial robot's CoM) in the derivation of flatness maps [see (9)] to improve the accuracy of the planned RPMs, due to the probable underestimation of the payload's mass and the closed-loop control law's need to regulate inevitable tracking errors (in position, velocity, attitude, and body rate), the required RPMs may always exceed the planned RPMs, making the planned trajectory too agile for the MARTS to track. Therefore, our schemes introduce robustification mechanisms for both the planner and the controller to deal with the above issues and ensure safe tracking based on the following three strategies:

1) Set the payload's mass provided to the planner as the estimated payload's mass plus the statistical error bound of the estimation scheme [see (48)].
2) Leave a fixed margin for the RPM away from the maximum RPM that the battery can tolerate to reduce the possibility of RPM saturation caused by inevitable tracking errors.
3) When the desired RPM required by the real-time control exceeds the maximum admissible RPM, a module for saturation on RPMs is necessary. Following the principle of prioritizing the satisfaction of control torque, the desired thrust is reduced so that the reduced desired RPM is less than or equal to the maximum admissible RPM.

VII. Benchmark, Ablation Study, and Simulation

In this section, benchmarks, ablation studies, and simulations are carried out to validate our proposed trajectory planning scheme for the SAAT in complex environments. First, we compare our planning scheme with a SOTA planner and analyze the performance in terms of responsiveness and trajectory quality (see Section VII-A). Then, we conduct ablation studies (see Section VII-B) to validate the necessity of some design choices employed in our planning scheme. Finally, we give a simulation of trajectory replanning when the MARTS experiences a single aerial robot's failure (see Section VII-C).

The optimization and visualization of the trajectories are implemented on a laptop with an Intel Core i5-10300H CPU(2.5 GHz) and do not depend on any hardware acceleration. The code implementation of our methods is based on C++11. The source code of our framework is available online.[1] Important parameters used for the following benchmarks, simulations in this section, and real-world experiments in Section VIII are shown in Table I.

[1] [Online]. Available: https://github.com/WwwangJJ/MARTS-Planner-ROS

TABLE I
PARAMETER SETTING FOR SIMULATIONS AND EXPERIMENTS

System Parameter					
m	320g	l	1.2m		
$\boldsymbol{\zeta}$	$(0, 0, -4.8cm)$				
$\mathbf{J}$	$diag(4.463, 4.725, 5.340) \times 10^{-4} kg \cdot m^2$				
Planning Parameter					
d_{oa}^L	$0.2m$	d_{oa}^Q	$0.3m$	d_{oa}^c	$0.2m$
K	7	d_s	0.2m	r_{max}	$24000RPM$
λ_{rpm}	10^5	F_{min}	$0.24N$	F_{max}	$2.4N$
λ_T	2000.0	λ_z	0.3	λ_{oa}	10^4
λ_{ra}	10^4	λ_d	10^7		
Control Parameter					
cof	20Hz	W	1000		
K_p	$diag(12.0, 12.0, 3.0)$	K_v	$diag(4.0, 4.0, 2.0)$		
$K_{\boldsymbol{\Theta}}$	$diag(70.0, 100.0, 19.0)$	$K_{\boldsymbol{\omega}}$	$diag(10.0, 12.0, 3.0)$		
K_I	$diag(0.0, 0.0, 0.3)$				

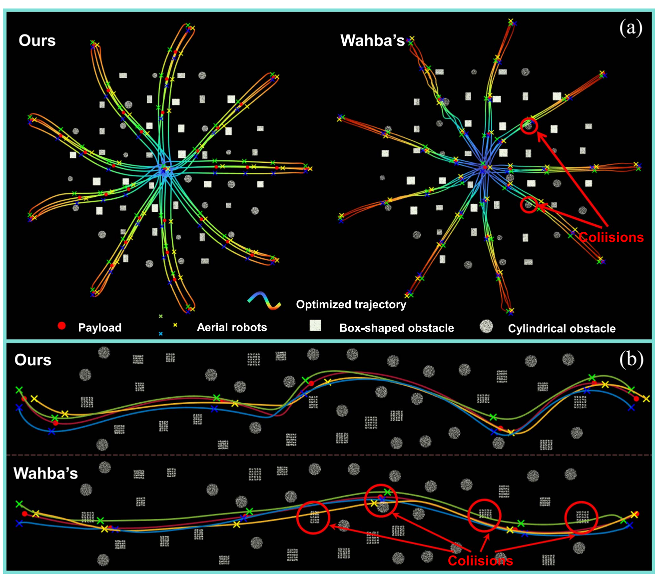


Fig. 6. Comparison of the trajectories planned, respectively, by our method and Wahba's method.

A. Benchmark Comparisons

In this work, a SOTA kinodynamic motion planning scheme for the payload transportation using MARTS proposed by Wahba et al. [38] is selected for benchmark comparisons. Wahba's scheme includes a front-end geometric path planner based on the RRT* algorithm and a back-end nonlinear trajectory planner based on the differential dynamic programming [50], both of which rely on the Flexible Collision Library [51] for collision checking.

The first experiment tests the qualities of trajectories planned by the two methods. We randomly generate cylindrical and square obstacles of different densities in a circular area with a radius of 18 m. The sparse, medium, and dense densities correspond to 3.2, 2.6, and 2.0 m average obstacle spacing, respectively, among which the medium-density environment is shown in Fig. 6(a). We select the center of the circular environment as the starting point and uniformly choose 36 points on a circle with a radius of 21 m as the target points to, respectively, plan 36 trajectories using the two methods. The bounds of constraints considered in both methods are set to be the same, as listed in Table I. An optimized result is successful if the planner generates a collision-free trajectory. For each obstacle density, we evaluate the two methods for the MARTS consisting of three aerial robots in terms of the success rate of these 36 trajectories.

TABLE II
PERFORMANCE COMPARISON BETWEEN THE TWO METHODS

Scenario	Method	success rate(%)
Sparse	Wahba's	77.8
	Ours	**100**
Medium	Wahba's	61.1
	Ours	**100**
Dense	Wahba's	38.9
	Ours	**100**

The result is shown in Table II. As the density of obstacles increases, the success rate of Wahba's scheme significantly decreases, whereas our scheme maintains a high success rate. This is due to the fact that Wahba's scheme transforms both the terminal constraint and the collision avoidance constraint into soft penalty functions. When the trajectory has a complex topology, the collision avoidance constraint may be coupled with the terminal constraint, resulting in conflicting gradient directions. This will cause Wahba's trajectory to fall into a local optimum, resulting in the collision avoidance constraint not being satisfied. In contrast, our trajectory always satisfies the terminal constraint, and the excellent spatiotemporal deformation ability of $\mathcal{C}_{\mathrm{MINCO}}$ is more helpful to avoid obstacles successfully. Compared to Wahba's scheme, our scheme can generate smoother trajectories. Besides, we further construct an environment as shown in Fig. 6(b), where the average spacing between obstacles is 1.6 m. We utilize the two schemes to plan trajectories between start and target points that are 36 m apart. It can be seen that our scheme can fully utilize the space to deform the trajectory so that the trajectory goes around the obstacle in a silky smooth manner, unlike Wahba's trajectory, which is always close to the edge of the obstacle due to the potential conflict between penalty functions.

The second experiment tests the responsiveness for both the front-end path planner and the back-end trajectory planner of the two schemes when the MARTS contains different numbers of aerial robots. This experiment is conducted in the medium-density environment. For both schemes, the number of decision variables and constraints scales linearly with the number of aerial robots. Nevertheless, whether the front-end path planning or the back-end trajectory planning, our scheme reduces the time consumption by two orders of magnitude compared to Wahba's scheme, and the curves of time consumptions for two path planners and two trajectory planners as the number of aerial robots grows are shown in Fig. 7. The reasons why our path planner outperforms Wahba's geometric path planner in efficiency are analyzed as follows. To reduce sampling complexity, Wahba et al. rely on a preprocessing to generate the configuration set required for sampling, which consumes some computational

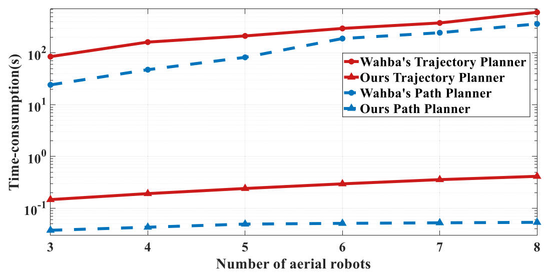


Fig. 7. Responsiveness comparison of the two schemes.

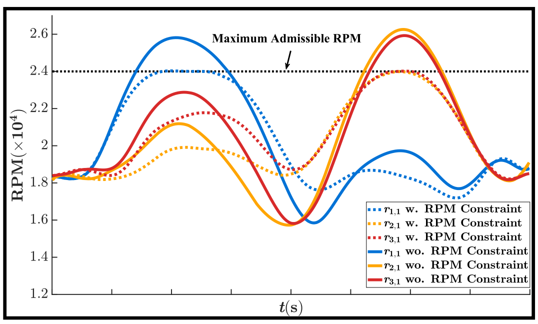


Fig. 8. Result of the ablation study for the maximum RPM constraint.

time. Besides, they employ a complex energy cost function to smooth paths, which partially replaces the functionality of the back-end trajectory planner. This results in higher computational complexity compared to path planners utilizing distance cost functions. In contrast, our system-level path planner employs a simple distance-based heuristic function, delegating the entire trajectory smoothing to our highly efficient back-end trajectory planner. This design enables our path planner to achieve superior path search efficiency. For two back-end trajectory planners, theoretically, the time consumption of Wahba's trajectory planner grows cubically, whereas our trajectory planner exhibits approximately linear growth. The reason for the improved responsiveness is that in each iteration of optimization, as the number of aerial robots increases, the elapsed time for constraint evaluation and penalty functions calculation grows linearly and the elapsed time for trajectory generation ($\boldsymbol{\varpi}^*, \boldsymbol{\tau}^* \rightarrow \mathbf{c}, \mathbf{T}$, as shown in Fig. 2) hardly increases.

B. Ablation Study

1) Scenario 1—Ablation Study for Maximum RPM Constraint: We first optimize a straight-line trajectory for the MARTS to verify the necessity of the maximum RPM constraint designed in Section IV-D. The partial planned RPMs of the three aerial robots with and without this constraint are shown in Fig. 8. It can be seen that this constraint successfully limits the motor's RPM to within the maximum admissible RPM.

2) Scenario 2—Ablation Study for ζ_n: Then, we further plan a new agile trajectory to verify the effect of ζ_n on the accuracy of planned RPMs. Both RPMs neglecting ζ_n ($\zeta_n = 0$) and RPMs using the real ζ_n ($\zeta_n = -4.8$ cm) of this trajectory are shown in Fig. 9, and the motors selected, respectively, for the three aerial robots are those whose RPMs are most underestimated when ζ_n is ignored. The maximum relative errors 12.3%, 19.2%, and

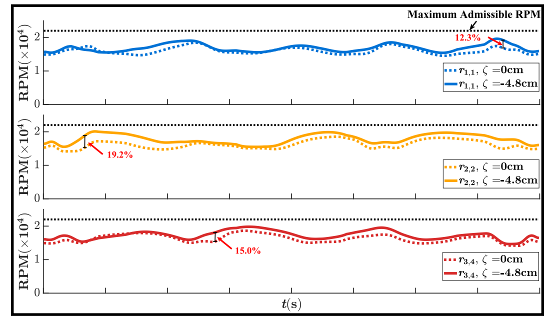


Fig. 9. Result of the ablation study for ζ_n.

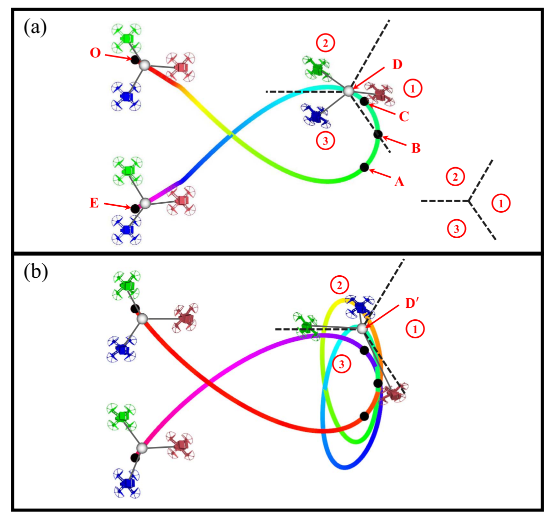


Fig. 10. Ablation study for the cable's vectorial constraints defined in (35). The trajectories optimized with and without these constraints are shown in (a) and (b), respectively.

15.0% prove the importance of ζ_n in improving the accuracy of the planned results.

3) Scenario 3—Ablation Study for Cable's Vectorial Constraint: The cable's vectorial constraints defined in (35) separately restrict the aerial robots as well as their attached cables to three regions, as shown in the lower right corner of Fig. 10(a). In this experiment, points O and E are set as the start and target points of the payload's trajectory, respectively. A, B, and C are fixed waypoints that the payload's trajectory must pass through. As shown in Fig. 10, compared with the planning scheme using vectorial constraints, the planning scheme without using vectorial constraints falls into a local optimum and generates a trajectory with a more complex topology. A major difference between these two trajectories is that at point D' in Fig. 10(b), the relative arrangement of the red and blue aerial robots differs from the initial relative arrangement at the start point O and the final relative arrangement at the target point E, in contrast to the point D in Fig. 10(a), where the three aerial robots always maintain a relative arrangement as same as those at the start point O and the target point E. Therefore, the vectorial constraints restrict the magnitude of the trajectory deformation by reducing the number of possible combinations among the cables' force vectors, which greatly alleviates the local optimum.

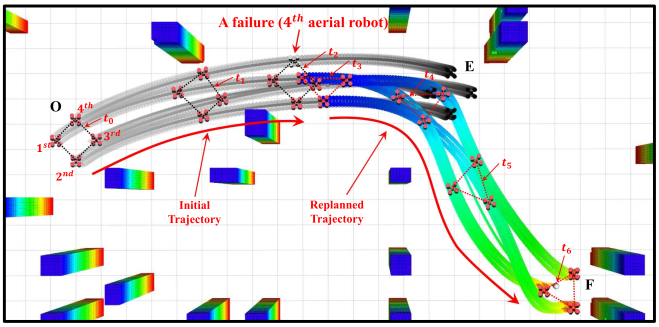


Fig. 11. Result of the trajectory replanning after the failure of a single aerial robot. The MARTSs, consisting of four aerial robots, are performing agile transportation along a trajectory targeting point E. At time stamp t_2, the fourth aerial robot encounters a failure. Then, our planner immediately replans a new trajectory targeting point F for the MARTS containing the remaining three aerial robots, and completes the trajectory switch at time stamp t_3.

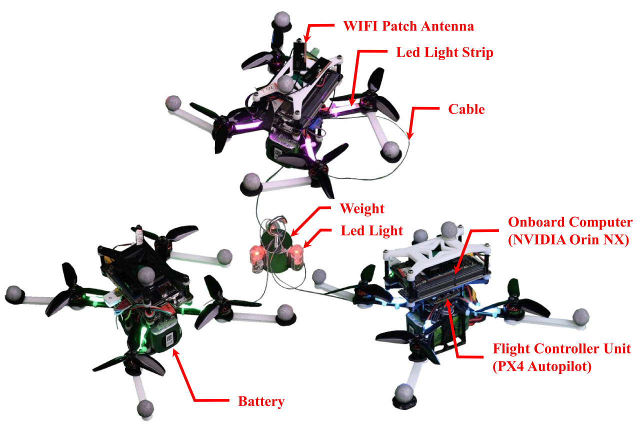


Fig. 12. Illustration of our MARTS consisting of three aerial robots.

C. Simulation

The failure of any single aerial robot in the MARTS can cause the MARTS to crash. Therefore, when the MARTS suffers from such a failure, the planner must possess the capability to quickly replan the trajectory for the incomplete MARTS consisting of the remaining aerial robots. In this section, we give a simulation for trajectory replanning after such a failure occurs based on the assumption that the failed aerial robot can recognize its failure in a very short time and automatically isolate itself from the MARTS. Replanning takes only 95 ms and the result is shown in Fig. 11 that the replanned trajectory ensures a smooth transition of the system state at the switching waypoint of two trajectories.

VIII. Real-World Experiments

A. System Configuration

To carry out practical experiments, we deploy two MARTSs, respectively, consisting of three aerial robots and five aerial robots. Each aerial robot weighs 320 g, has a thrust-to-weight ratio of about 3.0, and has a 140-mm wheelbase, as shown in Fig. 12. Each aerial robot is equipped with an NVIDIA Jetson Orin NX as the onboard computer, a Kakute H7 MINI as the flight control unit, and a Wi-Fi module. APM firmware is flashed into the flight control unit to get the motor's RPM. An EKF fusing the information from the motion capture system and the

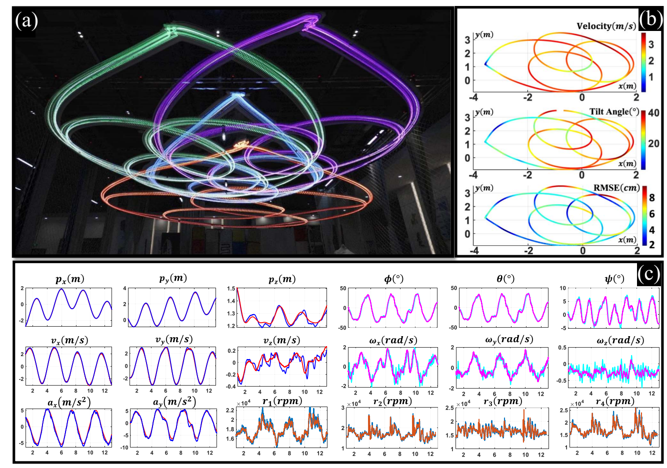


Fig. 13. Control results for tracking agile trajectory in free space, whose agility approaches the limit that the MARTS can successfully execute. (a) Long exposure image of the agile transportation trajectory, in which the red curve represents the payload's trajectory and the other three curves represent the three aerial robot's trajectories. (b) Instantaneous velocity, tilt angle, and RMSE of the second aerial robot displayed on its trajectory, respectively. (c) Result of trajectory tracking, including the position, velocity, acceleration, attitude, bodyrate, and motors' RPMs. The curves using colors , , and represent the desired states and the curves using colors , , and represent the actual state.

IMU is used to estimate the odometry for each aerial robot. The IMU frequency is set to 333 Hz. The software modules, including odometry estimation and flight control (both the outer-loop controller and inner-loop controller), run in real time at IMU's frequency on the NX computer. The optimized trajectory can be transmitted to each aerial robot through a broadcast network. For the experiments in complex environments, we use FAST-lio2, an excellent LiDAR-inertial odometry framework to get the point cloud data beforehand and then build the ESDF.

B. Control Scheme Validation for Agile Transportation

In this section, we construct several scenarios to validate the performance of our control scheme, and the details are as follows.

1) Scenario 1—Estimation for Payload's Mass: In this experiment, we select five payloads with different masses. Once the MARTS has lifted a payload off the ground and stabilized it at a predefined hovering position, the data from different time intervals can help to estimate the mass of each payload ten times by (48) in Section VI. By estimating a more accurate thrust coefficient for the motor in advance, the estimation accuracy of cables' tensions can be improved, ensuring that all the estimation errors do not exceed 4.2%, as listed in Fig. 14. The results show that our method can approximately estimate the mass of the payload.

2) Scenario 2—Agile Transportation to the Limit of Aerial Robot's Thrust: In this experiment, through regulating the temporal vector $\mathbf{T}$, we can generate a type of agile trajectories with different centripetal accelerations as shown in Fig. 13(a). Three weights weighing 100, 150, and 200 g, respectively, act as the payload. To evaluate the tracking errors of our control scheme, for each weight, we improve the acceleration of the trajectory as much as possible empirically until any motor's maximum

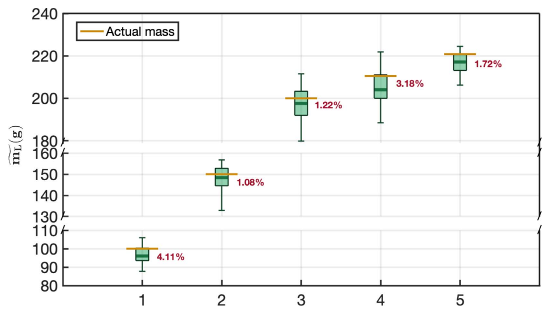


Fig. 14. Estimations for the masses of five different payloads.

TABLE III
MAXIMUM AGILITY TESTS OF THE MARTS FOR MULTIPLE WEIGHTS

$100g$	Max Vel(m/s)	4.4	4.5	4.55	4.6
	Max Acc(m/s^2)	9.104	9.126	9.175	9.184
	RMSE(cm)	**4.634**	**4.916**	**5.261**	**6.406**
$150g$	Max Vel(m/s)	4.0	4.1	4.15	4.2
	Max Acc(m/s^2)	7.278	7.859	7.942	8.443
	RMSE(cm)	**4.412**	**4.569**	**5.454**	**6.209**
$200g$	Max Vel(m/s)	3.6	3.7	3.75	3.8
	Max Acc(m/s^2)	5.749	6.126	6.292	6.458
	RMSE(cm)	**4.553**	**5.042**	**5.288**	**5.574**

TABLE IV
ABLATION STUDY OF THE FORCE COMPENSATION

	INDI		Reference Force	
Max Acc (m/s^2)	RMSE (cm)	MAXE (cm)	RMSE (cm)	MAXE (cm)
4.9	**3.910**	**7.141**	7.898	11.713
6.3	**4.129**	**7.449**	8.214	14.667
7.7	**4.383**	**9.315**	8.435	17.399
9.1	**4.674**	**10.267**	✗	✗

instantaneous RPM of any aerial robot in the MARTS along the trajectory approaches the maximum admissible 24 000 RPM that the battery can withstand (only 13 600 RPM is required for hovering without a payload). The maximum velocity (MAX Vel) and maximum acceleration (Max Acc) of the planned trajectory are recorded in Table III. Each trajectory is repeatedly executed five times to calculate the root-mean-square error (RMSE) of trajectory tracking. The results shown in Fig. 13(b) and (c) indicate that our control scheme can successfully track the agile trajectory up to the limit of the RPM that can be provided by the practical aerial robot in the MARTS.

3) Scenario 3—Comparative Study of Cable's Force Compensation: In this experiment, we replace the accurate force compensation based on INDI in the outer-loop control of the integrated control scheme with the reference force provided by the trajectory as a comparative control scheme. Four trajectories with different maximum accelerations are planned for this comparative study. Then, we transport a weight weighing 100 g along each of the four planned trajectories five times, respectively, using both the control schemes for a comparison. For each trajectory, each control scheme's maximum error (MAXE) along the trajectory and the RMSE are recorded in Table IV. For the control scheme using the reference force, the inevitable tracking

TABLE V
RMSES OF MULTIPLE PAYLOADS WITH UNCERTAINTIES

	$\pm 0\%$	$+10\%$ / -10%	$+20\%$ / -20%	$+30\%$ / -30%
Weight	4.472	4.563	4.747	4.897 (↑ 9.5%)
		4.804	4.867	5.158(↑ 15.3%)
Carton	4.935(↑ 10.4%)	5.243	5.424	5.493(↑ 22.8%)
		5.054	5.142	5.450(↑ 21.9%)
Bottle	5.039(↑ 12.7%)	5.227	5.386	5.575(↑ 24.7%)
		5.132	5.383	5.499(↑ 23.0%)

TABLE VI
RMSES OF TRANSPORTATIONS UNDER DIFFERENT CONDITIONS

	Benchmark	Low wind	Medium wind	Large wind	Comm Delay
RMSE (cm)	4.553	5.164	5.995	8.221	5.424

error of the payload induces the actual cable's force to periodically deviate from the reference force, leading to oscillation in the positions of aerial robots. This oscillation, in turn, further worsens the tracking precision of the payload. However, our control scheme can effectively avoid this oscillation since it accurately compensates for the actual cable's force. Therefore, this experiment validates the importance of the accurate force compensation based on INDI to substitute the reference force from the trajectory in the integrated control scheme for agile transportation.

4) Scenario 4—Robustness Against Payload's Uncertainties: In this experiment, we use a weight, a carton, and a water bottle respectively as the payload, all of which weigh 200 g. First, we use 200 g as the payload's mass to plan a trajectory, which is denoted by $\pm 0\%$. Then, we increase and decrease 200 g by 10%, 20%, and 30% as the payload's masses for the trajectory planning to simulate imprecise estimations for the payload's mass and plan six additional trajectories. We adjust the planning parameters to ensure that the maximum velocities of all these seven trajectories are 3.5 m/s. Each payload is transported by each trajectory three times to calculate the RMSE. For all these tests, the cable cannot be attached to the payload's CoM strictly, and none of these payloads can be simplified as mass points, which implies that there exists an unavoidable swing of the payload around the attaching point. All the payloads can be successfully transported by our MARTS using all these trajectories planned by the imprecise payload's masses, and RMSEs of these tests are recorded in Table V.

In the following two scenarios, we use the weight weighing 200 g as the payload and use the same trajectory as in Scenario 2 with a maximum velocity of 3.6 m/s and a maximum acceleration of 5.749 m/s^2 to further verify the robustness of the control scheme. Under each condition, this trajectory are executed repeatedly five times to calculate the RMSE.

5) Scenario 5—Robustness Against Wind Disturbances: We use a fan to simulate the wind disturbances and supplement experiments, respectively under low, medium, and large wind intensities. The results are shown in Table VI, where "Benchmark" represents experiments unaffected by wind disturbance and communication delay.

6) Scenario 6—Robustness Against Communication Delay: In this experiment, after planning the agile trajectory, we first immediately sent the trajectory to the first aerial robot, then sent the trajectory to the second and third aerial robots, respectively, after a 25-ms delay and after a 50-ms delay. Three aerial robots immediately execute trajectory tracking after receiving the trajectories. The result of the comparison between the RMSE under the condition of communication delays and the RMSE under the benchmark condition is shown in Table VI. At any given moment, communication delays cause the actual formation of all the aerial robots in the MARTS to differ from the reference formation provided by the trajectory, and the greater the instantaneous velocity of the trajectory, the larger the error between the actual formation and the reference formation. The formation error causes the actual tension of the cable to differ from the reference tension.

Reviewing results of scenarios 4–6, the reason for the robustness of our control scheme is that our control scheme compensates for the actual resultant external force and resultant torque in real time. Therefore, even if the actual resultant external force and resultant external torque differ from the planned force and torque, this control scheme can still effectively track the trajectory. Thus, this control scheme exhibits robustness against not only uncertainties on the payload but also wind disturbances and communication delays.

C. System Scheme Validation in Various Scenarios

We further construct a variety of scenarios to validate the performance of our MARTS, including the collision avoidance for agile transportation in complex environments, the agility of the trajectory in narrow spaces, and the responsiveness of the trajectory planning. The details are described as follows.

1) Continuous S-Shaped Turns: An experimental environment with multiple consecutive S-shaped turns is constructed, as shown in Fig. 1(b). In this experiment, a safe and agile trajectory of 24.43 m is planned just using 162 ms. The MARTS transports a payload of 200 g smoothly through these consecutive S-shaped turns and reaches the target point, without colliding with any obstacle. This experiment validates the capability of our planning scheme to generate a safe and agile trajectory in real-time.

2) Narrow Gap: We construct a narrow gap with the width of 1 m smaller than the initial size (1.6 m) of the MARTS. In this experiment, our planning scheme contracts the lateral size of the MARTS by optimizing all the cables' directions to ensure safe transportation through the narrow gap. As shown in Fig. 15, the result validates that our planning scheme can generate an agile trajectory by adjusting the configuration of MARTS to ensure the obstacle avoidance, which enhances its adaptability to narrow spaces. This experiment verifies the agility of the trajectory planned by our planning scheme.

3) Emergent Replanning Toward a New Target: In many missions, the responsiveness of replanning is important for mission efficiency. To validate the responsiveness of our planning scheme, we artificially construct an emergent replanning scenario, as shown in Fig. 16(a). This test consists of the following five phases.

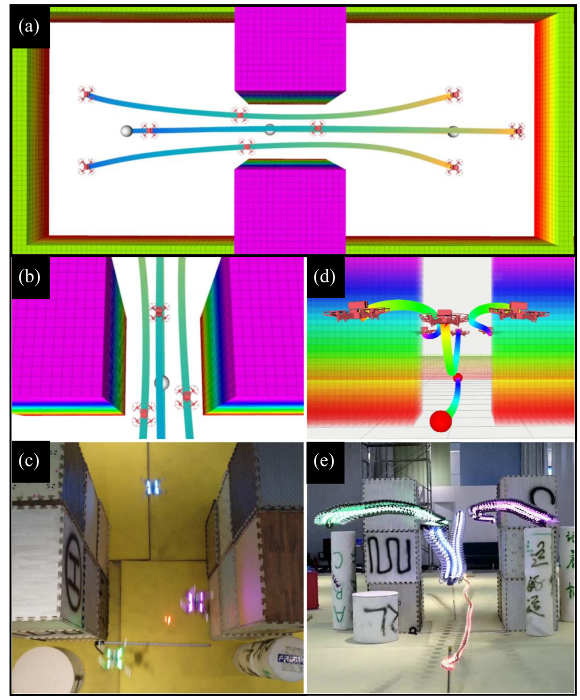


Fig. 15. Illustration of the MARTS passing through a narrow gap. (a) Simulation showing the whole optimized trajectory for safely passing through the gap by regulating the relative position among the aerial robots. (b) Top view in rviz at the moment of passing through the gap. (c) Snapshot w.r.t. the top view. (d) Side view of the optimized trajectory in rviz. (e) Long exposure image of the trajectory w.r.t. the side view.

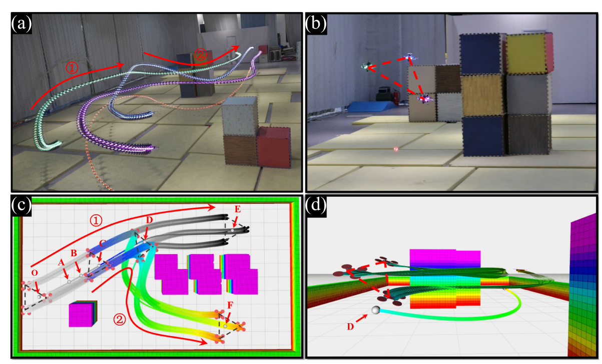


Fig. 16. Illustration of an emergent replanning for the MARTS toward a new target. (a) Long exposure image for the trajectories of the MARTS. (b) Simulation schematic for this emergency replanning. (c) Special zoomed-in view of the MARTS for the frame at point D. (d) Side view of the simulation schematic for this emergency replanning.

1) The trajectory $\widetilde{OE}$ is generated after the point E is selected as the target.
2) The MARTS flies along the trajectory $\widetilde{OE}$, and we select a new target point F for the MARTS when the payload reaches point A.
3) The planning module takes the state at point C, which is 200 ms ahead of point A, as the initial state and replans the trajectory.
4) The MARTS continually flies along the trajectory $\widetilde{AE}$ until it receives a new trajectory $\widetilde{CF}$ at the point B.

5) At point C, the MARTS switches the trajectory from trajectory $\widetilde{OE}$ to $\widetilde{CF}$.

Thanks to the real-time capability, our planning module just uses 72 ms to replan a new trajectory, which depresses the elapsed time required for replanning and thus strives for a larger safety distance away from the obstacle. Besides, an agile trajectory is generated by our planning scheme so that the MARTS can bypass the obstacle using a large maneuver and successfully switch the trajectory topology within the limited safety distance, which further improves the efficiency of the MARTS. Fig. 16(c) and (d) shows the snapshots of this maneuver. This experiment sufficiently verifies the responsiveness of our planning scheme. Moreover, we deploy a consecutive replanning experiment and present the result in our video.

4) Scalability to the Number of Aerial Robots: In this scenario, to validate the scalability of our planning and control scheme to the number of aerial robots contained in the MARTS, we integrate a MARTS consisting of five aerial robots and deploy an agile transportation experiment for a payload weighing 264 g, as shown in Fig. 1(c).

For more details about the benchmarks, ablations as well as simulations in Section VII, and real-world experiments in Section VIII, watch the video.

IX. Conclusion

In this article, we carefully derive the flatness maps for the aerial robot in the MARTS subject to dynamical coupling with payload and kinematic constraints provided by the cable. A real-time centralized trajectory planning scheme is proposed to generate safe, dynamically feasible, and agile trajectories for the MARTS in complex environments. A robust distributed control scheme is integrated to track agile trajectories without relying on the closed-loop control and state measurement for both the payload and cable, even if there exist uncertainties on the payload, communication delays, and wind disturbances. Finally, we deploy practical MARTSs containing different numbers of aerial robots with adequate simulations and experiments to validate the effectiveness of our planner and control schemes and exhibit great application value. It should be pointed out that there still exist limitations in our work. In this work, we only consider static obstacles and build the map beforehand, while ignoring the real-world experiment when any aerial robot in the MARTS encounters a failure. In the future, we will investigate methods for estimating the payload's pose and explore trajectory tracking control schemes based on these estimates. Besides, we will deploy real-time map updates for more complex dynamic obstacle scenarios and deploy a real-world experiment for the MARTS in the presence of a single-unit failure.

Acknowledgment

The authors would like to thank Jingao Liu, Qian Chen, and Che Qu at Beihang University and Shuhang Ji, Junxiao Lin, and Chen Yang at Zhejiang University for their strong support for this work.

Appendix

A. Derivation of Flatness Maps

Since the direction of $\mathbf{f}_n$ is parallel to $\mathbf{z}_\mathcal{B}^n$, we can obtain

$$\mathbf{z}_\mathcal{B}^n = \mathcal{N}(\mathbf{f}_n) \tag{49}$$

where $\mathcal{N}() : \mathbb{R}^3 \to \mathbb{R}^3$ is the vector unitization function defined as $\mathcal{N}(\mathbf{x}) \triangleq \mathbf{x}/\|\mathbf{x}\|_2, \mathbf{x} \in \mathbb{R}^3$. To recover the quaternion $\mathbf{q}_n$ w.r.t. $\mathbf{R}_n$ from $\mathbf{z}_\mathcal{B}^n$ and ψ_n, we use the rotation factorization known as Hopf fibration [45], which introduces the fewest singularities. The body frame $\mathbf{z}_\mathcal{B}^n$ is constructed through rotating the world frame $\mathcal{I}$ around its z-axis $\mathbf{z}_\mathcal{I}$ by ψ_n and continually rotating it around the current y-axis till its z-axis coincides with $\mathbf{z}_\mathcal{B}^n$, whose quaternion $\mathbf{q}_n$ can be computed as

$$\mathbf{q}_n = \mathbf{q}_{\mathbf{z}_\mathcal{B}^n} \odot \mathbf{q}_{\psi_n} \tag{50}$$

where $\odot$ is the quaternion multiplication, and $\mathbf{q}_\psi^i$ and $\mathbf{q}_{\mathbf{z}_\mathcal{B}^n}$ can be expanded as

$$\mathbf{q}_{\psi_n} = \left(\cos\frac{\psi_n}{2}, 0, 0, \sin\frac{\psi_n}{2}\right) \tag{51a}$$

$$\mathbf{q}_{\mathbf{z}_\mathcal{B}^n} = \frac{1}{\sqrt{2(\mathbf{z}_{\mathcal{B},3}^n + 1)}}\left(\mathbf{z}_{\mathcal{B},3}^n + 1, -\mathbf{z}_{\mathcal{B},2}^n, \mathbf{z}_{\mathcal{B},1}^n, 0\right). \tag{51b}$$

Thus, the body rate $\boldsymbol{\omega}_n$ can be calculated by

$$\boldsymbol{\omega}_n = 2\mathbf{q}_n^{-1} \odot \dot{\mathbf{q}}_n. \tag{52}$$

Then, substituting the inverse and derivative of $\mathbf{q}_n$ into (52), we obtain the expanded form of $\boldsymbol{\omega}_n$ as

$$\begin{aligned}\boldsymbol{\omega}_{n,1} &= \dot{\mathbf{z}}_{\mathcal{B},1}^n \sin(\psi_n) - \dot{\mathbf{z}}_{\mathcal{B},2}^n \cos(\psi_n) \\ &\quad - \frac{\dot{\mathbf{z}}_{\mathcal{B},3}^n\left[\mathbf{z}_{\mathcal{B},1}^n \sin(\psi_n) - \mathbf{z}_{\mathcal{B},2}^n \cos(\psi_n)\right]}{\mathbf{z}_{\mathcal{B},3}^n + 1}\end{aligned} \tag{53a}$$

$$\begin{aligned}\boldsymbol{\omega}_{n,2} &= \dot{\mathbf{z}}_{\mathcal{B},1}^n \cos(\psi_n) + \dot{\mathbf{z}}_{\mathcal{B},2}^n \sin(\psi_n) \\ &\quad - \frac{\dot{\mathbf{z}}_{\mathcal{B},3}^n\left[\mathbf{z}_{\mathcal{B},1}^n \cos(\psi_n) + \mathbf{z}_{\mathcal{B},2}^n \sin(\psi_n)\right]}{\mathbf{z}_{\mathcal{B},3}^n + 1}\end{aligned} \tag{53b}$$

$$\boldsymbol{\omega}_{n,3} = \frac{\mathbf{z}_{\mathcal{B},2}^n \dot{\mathbf{z}}_{\mathcal{B},1}^n - \mathbf{z}_{\mathcal{B},1}^n \dot{\mathbf{z}}_{\mathcal{B},2}^n}{\mathbf{z}_{\mathcal{B},3}^n + 1} + \dot{\psi}_n. \tag{53c}$$

Finally, (52) can be further differentiated to obtain the expanded form of $\dot{\boldsymbol{\omega}}_n$ as

$$\begin{aligned}\dot{\boldsymbol{\omega}}_{n,1} &= \ddot{\mathbf{z}}_{\mathcal{B},1}^n \sin(\psi_n) - \ddot{\mathbf{z}}_{\mathcal{B},2}^n \cos(\psi_n) \\ &\quad + \dot{\mathbf{z}}_{\mathcal{B},1}^n \cos(\psi_n)\dot{\psi}_n + \dot{\mathbf{z}}_{\mathcal{B},2}^n \sin(\psi_n)\dot{\psi}_n \\ &\quad - \sin(\psi_n)\frac{\ddot{\mathbf{z}}_{\mathcal{B},3}^n \mathbf{z}_{\mathcal{B},1}^n + \dot{\mathbf{z}}_{\mathcal{B},3}^n \dot{\mathbf{z}}_{\mathcal{B},1}^n + \dot{\mathbf{z}}_{\mathcal{B},3}^n \mathbf{z}_{\mathcal{B},2}^n \dot{\psi}_n}{\mathbf{z}_{\mathcal{B},3}^n + 1} \\ &\quad + \cos(\psi_n)\frac{\ddot{\mathbf{z}}_{\mathcal{B},3}^n \mathbf{z}_{\mathcal{B},2}^n + \dot{\mathbf{z}}_{\mathcal{B},3}^n \dot{\mathbf{z}}_{\mathcal{B},2}^n - \dot{\mathbf{z}}_{\mathcal{B},3}^n \mathbf{z}_{\mathcal{B},1}^n \dot{\psi}_n}{\mathbf{z}_{\mathcal{B},3}^n + 1} \\ &\quad + \frac{(\dot{\mathbf{z}}_{\mathcal{B},3}^n)^2\left[\mathbf{z}_{\mathcal{B},1}^n \sin(\psi_n) - \mathbf{z}_{\mathcal{B},2}^n \cos(\psi_n)\right]}{(\mathbf{z}_{\mathcal{B},3}^n + 1)^2}\end{aligned} \tag{54a}$$

$$\begin{aligned}\dot{\boldsymbol{\omega}}_{n,2} &= \ddot{\mathbf{z}}_{\mathcal{B},1}^n \cos(\psi_n) + \ddot{\mathbf{z}}_{\mathcal{B},2}^n \sin(\psi_n) \\ &\quad - \dot{\mathbf{z}}_{\mathcal{B},1}^n \sin(\psi_n)\dot{\psi}_n + \dot{\mathbf{z}}_{\mathcal{B},2}^n \cos(\psi_n)\dot{\psi}_n \\ &\quad - \cos(\psi_n)\frac{\ddot{\mathbf{z}}_{\mathcal{B},3}^n \mathbf{z}_{\mathcal{B},1}^n + \dot{\mathbf{z}}_{\mathcal{B},3}^n \dot{\mathbf{z}}_{\mathcal{B},1}^n + \dot{\mathbf{z}}_{\mathcal{B},3}^n \mathbf{z}_{\mathcal{B},2}^n \dot{\psi}_n}{\mathbf{z}_{\mathcal{B},3}^n + 1}\end{aligned}$$

$$- \sin(\psi_n) \frac{\ddot{\mathbf{z}}^n_{\mathcal{B},3}\mathbf{z}^n_{\mathcal{B},2} + \dot{\mathbf{z}}^n_{\mathcal{B},3}\dot{\mathbf{z}}^n_{\mathcal{B},2} - \dot{\mathbf{z}}^n_{\mathcal{B},3}\mathbf{z}^n_{\mathcal{B},1}\dot{\psi}_n}{\mathbf{z}^n_{\mathcal{B},3}+1} + \frac{(\dot{\mathbf{z}}^n_{\mathcal{B},3})^2\left[\mathbf{z}^n_{\mathcal{B},1}\cos(\psi_n) + \mathbf{z}^n_{\mathcal{B},2}\sin(\psi_n)\right]}{(\mathbf{z}^n_{\mathcal{B},3}+1)^2} \tag{54b}$$

$$\dot{\boldsymbol{\omega}}_{n,3} = \frac{\mathbf{z}^n_{\mathcal{B},2}\ddot{\mathbf{z}}^n_{\mathcal{B},1} - \mathbf{z}^n_{\mathcal{B},1}\ddot{\mathbf{z}}^n_{\mathcal{B},2}}{\mathbf{z}^n_{\mathcal{B},3}+1} - \dot{\mathbf{z}}^n_{\mathcal{B},3}\frac{\mathbf{z}^n_{\mathcal{B},2}\dot{\mathbf{z}}^n_{\mathcal{B},1} - \mathbf{z}^n_{\mathcal{B},1}\dot{\mathbf{z}}^n_{\mathcal{B},2}}{(\mathbf{z}^n_{\mathcal{B},3}+1)^2} + \ddot{\psi}_n \tag{54c}$$

where $\dot{\mathbf{z}}^n_{\mathcal{B}}$ and $\ddot{\mathbf{z}}^n_{\mathcal{B}}$ can be, respectively, derived as

$$\dot{\mathbf{z}}^n_{\mathcal{B}} = \frac{\dot{\mathbf{f}}_n}{\|\mathbf{f}_n\|_2} - \frac{(\mathbf{f}_n^T\dot{\mathbf{f}}_n)\mathbf{f}_n}{\|\mathbf{f}_n\|_2^3} \tag{55a}$$

$$\ddot{\mathbf{z}}^n_{\mathcal{B}} = \frac{\ddot{\mathbf{f}}_n}{\|\mathbf{f}_n\|_2} - \frac{(\dot{\mathbf{f}}_n^T\dot{\mathbf{f}}_n + \mathbf{f}_n^T\ddot{\mathbf{f}}_n)\mathbf{f}_n + 2(\mathbf{f}_n^T\dot{\mathbf{f}}_n)\dot{\mathbf{f}}_n}{\|\mathbf{f}_n\|_2^3} + \frac{3(\mathbf{f}_n^T\dot{\mathbf{f}}_n)^2\mathbf{f}_n}{\|\mathbf{f}_n\|_2^5} \tag{55b}$$

and the first and second derivatives of $\mathbf{f}_n$ required in (55) can be derived as

$$\dot{\mathbf{f}}_n = \dddot{\mathbf{p}} + l\dddot{\boldsymbol{\rho}}_n + \frac{1}{m}\left(\dot{F}_n\boldsymbol{\rho}_n + F_n\dot{\boldsymbol{\rho}}_n\right) \tag{56a}$$

$$\ddot{\mathbf{f}}_n = \mathbf{p}^{(4)} + l\boldsymbol{\rho}_n^{(4)} + \frac{1}{m}\left(\ddot{F}_n\boldsymbol{\rho}_n + 2\dot{F}_n\dot{\boldsymbol{\rho}}_n + F_n\ddot{\boldsymbol{\rho}}_n\right). \tag{56b}$$

REFERENCES

[1] K. Sreenath, T. Lee, and V. Kumar, "Geometric control and differential flatness of a quadrotor UAV with a cable-suspended load," in *Proc. 52nd IEEE Conf. Decis. Control*, 2013, pp. 2269–2274.

[2] P. Foehn et al., "Fast trajectory optimization for agile quadrotor maneuvers with a cable-suspended payload," *Robot.: Sci. Syst.*, Jul. 2017, doi: 10.15607/RSS.2017.XIII.0302017.

[3] S. Tang, V. Wüest, and V. Kumar, "Aggressive flight with suspended payloads using vision-based control," *IEEE Robot. Automat. Lett.*, vol. 3, no. 2, pp. 1152–1159, Apr. 2018.

[4] C. Y. Son, H. Seo, D. Jang, and H. J. Kim, "Real-time optimal trajectory generation and control of a multi-rotor with a suspended load for obstacle avoidance," *IEEE Robot. Automat. Lett.*, vol. 5, no. 2, pp. 1915–1922, Apr. 2020.

[5] J. Zeng, P. Kotaru, M. W. Mueller, and K. Sreenath, "Differential flatness based path planning with direct collocation on hybrid modes for a quadrotor with a cable-suspended payload," *IEEE Robot. Automat. Lett.*, vol. 5, no. 2, pp. 3074–3081, Apr. 2020.

[6] S. Belkhale, R. Li, G. Kahn, R. McAllister, R. Calandra, and S. Levine, "Model-based meta-reinforcement learning for flight with suspended payloads," *IEEE Robot. Automat. Lett.*, vol. 6, no. 2, pp. 1471–1478, Apr. 2021.

[7] G. Yu, D. Cabecinhas, R. Cunha, and C. Silvestre, "Aggressive maneuvers for a quadrotor-slung-load system through fast trajectory generation and tracking," *Auton. Robots*, vol. 46, no. 4, pp. 499–513, 2022.

[8] H. Li, H. Wang, C. Feng, F. Gao, B. Zhou, and S. Shen, "AutoTrans: A complete planning and control framework for autonomous UAV payload transportation," *IEEE Robot. Automat. Lett.*, vol. 8, no. 10, pp. 6859–6866, Oct. 2023.

[9] H. Wang, H. Li, B. Zhou, F. Gao, and S. Shen, "Impact-aware planning and control for aerial robots with suspended payloads," *IEEE Trans. Robot.*, vol. 40, pp. 2478–2497, 2024.

[10] D. Mellinger, M. Shomin, N. Michael, and V. Kumar, "Cooperative grasping and transport using multiple quadrotors," in *Proc. 10th Int. Symp. Distrib. Auton. Robotic Syst.*, 2013, pp. 545–558.

[11] G. Loianno and V. Kumar, "Cooperative transportation using small quadrotors using monocular vision and inertial sensing," *IEEE Robot. Automat. Lett.*, vol. 3, no. 2, pp. 680–687, Apr. 2018.

[12] Z. Wang, S. Singh, M. Pavone, and M. Schwager, "Cooperative object transport in 3D with multiple quadrotors using no peer communication," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2018, pp. 1064–1071.

[13] M. Bernard, K. Kondak, I. Maza, and A. Ollero, "Autonomous transportation and deployment with aerial robots for search and rescue missions," *J. Field Robot.*, vol. 28, no. 6, pp. 914–931, 2011.

[14] C. Masone, H. H. Bülthoff, and P. Stegagno, "Cooperative transportation of a payload using quadrotors: A reconfigurable cable-driven parallel robot," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2016, pp. 1623–1630.

[15] P. Prajapati, S. Parekh, and V. Vashista, "On the human control of a multiple quadcopters with a cable-suspended payload system," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2020, pp. 2253–2258.

[16] G. Li, R. Ge, and G. Loianno, "Cooperative transportation of cable suspended payloads with MAVs using monocular vision and inertial sensing," *IEEE Robot. Automat. Lett.*, vol. 6, no. 3, pp. 5316–5323, Jul. 2021.

[17] J. Wehbeh, S. Rahman, and I. Sharf, "Distributed model predictive control for UAVs collaborative payload transport," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2020, pp. 11666–11672.

[18] E. Kaufmann, L. Bauersfeld, A. Loquercio, M. Müller, V. Koltun, and D. Scaramuzza, "Champion-level drone racing using deep reinforcement learning," *Nature*, vol. 620, no. 7976, pp. 982–987, 2023.

[19] M. Gassner, T. Cieslewski, and D. Scaramuzza, "Dynamic collaboration without communication: Vision-based cable-suspended load transport with two quadrotors," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2017, pp. 5196–5202.

[20] M. Tognon, C. Gabellieri, L. Pallottino, and A. Franchi, "Aerial co-manipulation with cables: The role of internal force for equilibria, stability, and passivity," *IEEE Robot. Automat. Lett.*, vol. 3, no. 3, pp. 2577–2583, Jul. 2018.

[21] A. Tagliabue, M. Kamel, R. Siegwart, and J. Nieto, "Robust collaborative object transportation using multiple MAVs," *Int. J. Robot. Res.*, vol. 38, no. 9, pp. 1020–1044, 2019.

[22] T. Lee, K. Sreenath, and V. Kumar, "Geometric control of cooperating multiple quadrotor UAVs with a suspended payload," in *Proc. 52nd IEEE Conf. Decis. Control*, 2013, pp. 5510–5515.

[23] T. Lee, "Geometric control of quadrotor UAVs transporting a cable-suspended rigid body," *IEEE Trans. Control Syst. Technol.*, vol. 26, no. 1, pp. 255–264, Jan. 2018.

[24] G. Li, X. Liu, and G. Loianno, "RotorTM: A flexible simulator for aerial transportation and manipulation," *IEEE Trans. Robot.*, vol. 40, pp. 831–850, 2024.

[25] J. Geng and J. W. Langelaan, "Cooperative transport of a slung load using load-leading control," *J. Guid., Control, Dyn.*, vol. 43, no. 7, pp. 1313–1331, 2020.

[26] G. Li and G. Loianno, "Nonlinear model predictive control for cooperative transportation and manipulation of cable suspended payloads with multiple quadrotors," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2023, pp. 5034–5041.

[27] K. Wahba and W. Hönig, "Efficient optimization-based cable force allocation for geometric control of a multirotor team transporting a payload," *IEEE Robot. Automat. Lett.*, vol. 9, no. 4, pp. 3688–3695, Apr. 2024.

[28] K. Klausen, C. Meissen, T. I. Fossen, M. Arcak, and T. A. Johansen, "Cooperative control for multirotors transporting an unknown suspended load under environmental disturbances," *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 2, pp. 653–660, Mar. 2020.

[29] H. G. De Marina and E. Smeur, "Flexible collaborative transportation by a team of rotorcraft," in *Proc. Int. Conf. Robot. Automat.*, 2019, pp. 1074–1080.

[30] N. Michael, J. Fink, and V. Kumar, "Cooperative manipulation and transportation with aerial robots," *Auton. Robots*, vol. 30, pp. 73–86, 2011.

[31] J. Fink, N. Michael, S. Kim, and V. Kumar, "Planning and control for cooperative manipulation and transportation with aerial robots," *Int. J. Robot. Res.*, vol. 30, no. 3, pp. 324–334, 2011.

[32] Q. Jiang and V. Kumar, "The inverse kinematics of cooperative transport with multiple aerial robots," *IEEE Trans. Robot.*, vol. 29, no. 1, pp. 136–145, Feb. 2013.

[33] D. Sanalitro, H. J. Savino, M. Tognon, J. Cortés, and A. Franchi, "Full-pose manipulation control of a cable-suspended load with multiple UAVs under uncertainties," *IEEE Robot. Automat. Lett.*, vol. 5, no. 2, pp. 2185–2191, Jan. 2020.

[34] M. Manubens, D. Devaurs, L. Ros, and J. Cortés, "Motion planning for 6-D manipulation with aerial towed-cable systems," in *Proc. Proc. Robot.: Sci. Syst.*, Berlin, Germany, Jun. 2013, p. 8.

[35] J. Geng, P. Singla, and J. W. Langelaan, "Load-distribution-based trajectory planning and control for a multilift system," *J. Aerosp. Inf. Syst.*, vol. 19, no. 5, pp. 366–381, 2022.

[36] B. E. Jackson, T. A. Howell, K. Shah, M. Schwager, and Z. Manchester, "Scalable cooperative transport of cable-suspended loads with UAVs using distributed trajectory optimization," *IEEE Robot. Automat. Lett.*, vol. 5, no. 2, pp. 3368–3374, Mar. 2020.
[37] S. Sun and A. Franchi, "Nonlinear MPC for full-pose manipulation of a cable-suspended load using multiple UAVs," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2023, pp. 969–975.
[38] K. Wahba, J. Ortiz de Haro, M. Toussaint, and W. Hönig, "Kinodynamic motion planning for a team of multirotors transporting a cable-suspended payload in cluttered environments," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2023, pp. 12750–12757.
[39] K. Sreenath and V. R. Kumar, "Dynamics, control and planning for cooperative manipulation of payloads suspended by cables from multiple quadrotor robots," in *Proc. Robot.: Sci. Syst.*, 2013.
[40] D. Mellinger and V. Kumar, "Minimum snap trajectory generation and control for quadrotors," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2011, pp. 2520–2525.
[41] W. Xu, Y. Cai, D. He, J. Lin, and F. Zhang, "FAST-LIO2: Fast direct LiDAR-inertial odometry," *IEEE Trans. Robot.*, vol. 38, no. 4, pp. 2053–2073, Jan. 2022.
[42] E. Tal and S. Karaman, "Accurate tracking of aggressive quadrotor trajectories using incremental nonlinear dynamic inversion and differential flatness," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 3, pp. 1203–1218, Mar. 2021.
[43] X. Zhou et al., "Swarm of micro flying robots in the wild," *Sci. Robot.*, vol. 7, no. 66, 2022, Art. no. eabm 5954.
[44] B. Zhou, F. Gao, L. Wang, C. Liu, and S. Shen, "Robust and efficient quadrotor trajectory generation for fast autonomous flight," *IEEE Robot. Automat. Lett.*, vol. 4, no. 4, pp. 3529–3536, 2019.
[45] M. Watterson and V. Kumar, "Control of quadrotors using the Hopf fibration on SO (3)," in *Proc. 18th Int. Symp. Robot. Res.*, 2019, pp. 199–215.
[46] Z. Wang, X. Zhou, C. Xu, and F. Gao, "Geometrically constrained trajectory optimization for multicopters," *IEEE Trans. Robot.*, vol. 38, no. 5, pp. 3259–3278, Nov. 2022.
[47] F. Gao, L. Wang, B. Zhou, X. Zhou, J. Pan, and S. Shen, "Teach-repeat-replan: A complete and robust system for aggressive flight in complex environments," *IEEE Trans. Robot.*, vol. 36, no. 5, pp. 1526–1545, Oct. 2020.
[48] L. S. Jennings and K. L. Teo, "A computational algorithm for functional inequality constrained optimization problems," *Automatica*, vol. 26, no. 2, pp. 371–375, 1990.
[49] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, 3rd ed. Cambridge, MA, USA: Cambridge Univ. Press, 2007.
[50] J. Ortiz-Haro, W. Hönig, V. N. Hartmann, and M. Toussaint, "iDb-A∗: Iterative search and optimization for optimal kinodynamic motion planning," *IEEE Trans. Robot.*, vol. 41, pp. 2031–2049, 2025.
[51] J. Pan, S. Chitta, and D. Manocha, "FCL: A general purpose library for collision and proximity queries," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2012, pp. 3859–3866.

Yongchao Wang received the Ph.D. degree in aircraft design from Beihang University, Beijing, China, in 2021.

He is currently a Postdoctoral Fellow with the School of Aeronautic Science and Engineering, Beihang University. His research interests include the trajectory planning and control for the multiple quadcopter aerial transportation system, multi-autonomous-aerial-vehicle cooperative decision making, and reinforcement learning.

Junjie Wang received the bachelor's degree in mechanical engineering in 2024 from Zhejiang University, Hangzhou, China, where he is currently working toward the master's degree in control science and engineering with the FAST Lab, under the supervision of Prof. Fei Gao.

His research interests include planning and control for aerial robotics.

Xiaobin Zhou (Member, IEEE) received the B.S. degree from Northeastern University, Shenyang, China, in 2015, and the Ph.D. degree from Hunan University, Changsha, China, in 2021, both in mechanical engineering.

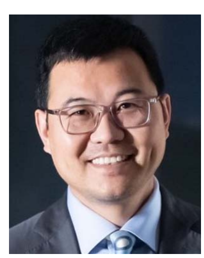
From 2018 to 2021, he was a Visiting Student with Concordia University, Montreal, QC, Canada, and Beihang University, Beijing, China. From 2021 to 2022, he was an Engineer with XPeng AeroHT. From 2022 to 2025, he was a Postdoctoral Fellow with Zhejiang University. He is currently an Assistant Professor with the School of Robotics and Automation, Nanjing University, Suzhou, China. His research interests include aerial robot design, autonomous navigation, and multirobot systems.

Tiankai Yang received the bachelor's degree in detection, guidance, and control technology from Northwestern Polytechnical University, Xi'an, China, in 2019. He is currently working toward the Ph.D. degree in control science and engineering with the FAST Lab, Zhejiang University, Hangzhou, China, under the supervision of Prof. Fei Gao.

His research interests include control and navigation for aerial robotics and unmanned vehicles.

Xin Zhou received the Ph.D. degree in electronic information from Zhejiang University, Hangzhou, China, in 2024.

He was a Postdoctoral Fellow with the Hong Kong University of Science and Technology, Hong Kong. He is currently the Chief Scientist Officer with Differential Robotics (Hangzhou) Technology Company, Ltd., Hangzhou. His research interests include vision language navigation for aerial robots, reinforcement-learning-based agile flight, and dexterous aerial manipulation.



Chao Xu (Senior Member, IEEE) received the Ph.D. degree in mechanical engineering from Lehigh University, Bethlehem, PA, USA, in 2010.

He is currently a Professor with the College of Control Science and Engineering, Zhejiang University, Hangzhou, China. He serves as the inaugural Dean of ZJU Huzhou Institute. He has authored or coauthored more than 100 papers in international journals, including *Science Robotics* and *Nature Machine Intelligence*. His research interests include flying robotics and control-theoretic learning. Dr. Xu is the Managing Editor for *IET Cyber-Systems & Robotics*.

Fei Gao (Member, IEEE) received the Ph.D. degree in electronic and computer engineering from the Hong Kong University of Science and Technology, Hong Kong, in 2019.

He is currently a tenured Associate Professor with the College of Control Science and Engineering, Zhejiang University, Hangzhou, China, where he leads the Aerial Embodied Intelligence Laboratory. He is also the founder of Differential Robotics Technology Company, Hangzhou. His research interests include aerial robots, autonomous navigation, motion planning, embodied intelligence, and swarm intelligence.